Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiments 8, 9

Effect of Attention Type and Attention Position in a Pretrained ResNet50

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
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1. Objective

The objective of this experiment is to understand how different attention mechanisms modify a pretrained convolutional neural network and how the **position at which attention is inserted** influences image classification performance.

A single pretrained architecture,

ResNet50 pretrained on ImageNet

is used throughout the experiment.

Students will investigate:

- channel attention;
- spatial attention;
- sequential channel–spatial attention;
- lightweight self-attention;
- different positions for inserting attention within ResNet50;
- single-stage and multi-stage attention;
- attention ordering;
- additional established attention mechanisms; and
- qualitative interpretation using internal attention maps and Grad-CAM.

The experiment emphasizes that the effectiveness of attention depends not only on the attention mechanism itself, but also on the characteristics of the feature representation on which it operates.

2. Learning Outcomes

After completing this experiment, students will be able to:

1. explain the stage-wise architecture of a pretrained ResNet50;
2. distinguish early, intermediate, and late convolutional features;
3. explain the motivation behind attention mechanisms;
4. implement channel attention;
5. implement spatial attention;
6. implement sequential channel–spatial attention;
7. implement a lightweight self-attention mechanism;
8. insert an attention module at different stages of a pretrained network;
9. analyze the influence of attention position;
10. compare different attention mechanisms under a controlled experimental protocol;
11. analyze classification accuracy and class-wise performance;
12. visualize internal attention maps;
13. generate Grad-CAM explanations;
14. distinguish an internal attention map from a post-hoc explanation map;
15. analyze accuracy–complexity trade-offs; and
16. perform controlled ablation studies.

3. Dataset

Recommended Dataset: TensorFlow Flowers (`tf_flowers`)

The TensorFlow Flowers dataset contains natural RGB flower images belonging to five classes:

`daisy`, `dandelion`, `roses`, `sunflowers`, `tulips`.

The images contain variations in:

- scale;
- viewpoint;
- illumination;
- background clutter;
- flower size;
- color;
- texture; and
- object location.

These variations make the dataset suitable for studying the behavior of attention mechanisms.

3.1 Image Preprocessing

Resize every input image to

$$224 \times 224 \times 3.$$

The image should be preprocessed using the preprocessing procedure expected by the ImageNet-pretrained ResNet50.
Suitable data augmentation may include:

- random horizontal flipping;
- small rotations;
- random zoom;
- random translation; and
- mild contrast variation.

The same augmentation strategy must be maintained for all model comparisons.

3.2 Dataset Split

Use the same training, validation, and testing split for every experiment:

$$\boxed{70\% : 15\% : 15\%}$$

for

$$\text{Training} : \text{Validation} : \text{Testing}.$$

The exact same samples should be retained in each subset for all ablation experiments.
This is necessary to ensure that differences in performance are caused by changes in the model rather than changes in the dataset split.

4. Pretrained Architecture: ResNet50

ResNet50 pretrained using ImageNet is used as the **only backbone architecture** in this experiment.
For an input image

$$I \in \mathbb{R}^{224\times224\times3},$$

the major ResNet50 stages produce approximately the following feature dimensions:

ResNet50 Stage	**Spatial Resolution**	**Channels**
Stem + MaxPool	56×56	64
conv2_x	56×56	256
conv3_x	28×28	512
conv4_x	14×14	1024
conv5_x	7×7	2048

As the network becomes deeper:

$$\boxed{\text{Spatial Resolution} \downarrow}$$

while

$$\boxed{\text{Feature Channels} \uparrow}$$

and the representation generally progresses from low-level visual information toward higher-level semantic information.

5. Understanding ResNet50 Feature Hierarchy

The approximate interpretation of different stages is:

- **Early features:** edges, colors, orientations, boundaries, and simple textures.
- **Intermediate features:** local patterns, repeated structures, flower parts, petal arrangements, and shape information.
- **Late features:** high-level object structures and class-discriminative semantic information.

Therefore, the same attention mechanism may produce different behavior when inserted at different locations.

6. Why Attention Position Matters

Attention is not independent of the features on which it operates.
Suppose an attention module receives

$$F \in \mathbb{R}^{H \times W \times C}.$$

The values of

$$H, \qquad W, \qquad C$$

and the semantic meaning of the feature tensor change significantly with network depth.
Therefore:

- **early-stage attention** operates on high-resolution, low-level visual information;
- **middle-stage attention** operates on intermediate features containing both spatial and semantic information; and
- **late-stage attention** operates on low-resolution, high-level semantic representations.

Thus, students should investigate both:

$$\boxed{\text{Which attention mechanism should be used?}}$$

and

$$\boxed{\text{Where should the attention mechanism be inserted?}}$$

7. Candidate Attention Positions

Define four candidate attention positions in ResNet50.

$$P_1: \text{ after conv2_x},$$

$$P_2: \text{ after conv3_x},$$

$$P_3: \text{ after conv4_x},$$

$$P_4: \text{ after conv5_x}.$$

The corresponding feature dimensions are:

Position	Location	Resolution	Channels
P_1	After conv2_x	56×56	256
P_2	After conv3_x	28×28	512
P_3	After conv4_x	14×14	1024
P_4	After conv5_x	7×7	2048

For the principal laboratory exercise, students should concentrate on

$$\boxed{P_2, \quad P_3, \quad P_4}$$

because P_1 contains comparatively high-resolution features and may substantially increase the computational cost of some attention mechanisms.

8. Baseline Model

The baseline is the pretrained ResNet50 without any additional attention mechanism.



The baseline establishes the reference performance against which the attention-based models are compared.

9. Channel Attention

Consider a feature tensor

$$F \in \mathbb{R}^{H \times W \times C}.$$

Channel attention determines the relative importance of different feature channels.
Global average pooling is first used to generate a channel descriptor:

$$z_c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} F(i, j, c).$$

Thus,

$$z \in \mathbb{R}^C.$$

The channel descriptor is transformed using two learnable transformations:

$$M_c(F) = \sigma\left(W_2 \delta\left(W_1 z\right)\right),$$

where:

- $\delta(\cdot)$ denotes a nonlinear activation such as ReLU;
- $\sigma(\cdot)$ denotes the sigmoid activation;
- W_1 reduces the channel representation; and
- W_2 restores the channel dimension.

The refined feature tensor is:

$$F_c = F \odot M_c(F).$$

Channel attention attempts to answer:

$$\boxed{\text{Which feature channels are important?}}$$

A reduction ratio of

$$r = 16$$

may be used as the default setting throughout the laboratory experiment.
The reduction ratio need not be treated as a separate mandatory ablation.

10. Spatial Attention

Spatial attention determines the spatial regions that should receive greater importance.
For the input feature tensor

$$F \in \mathbb{R}^{H\times W\times C},$$

compute average pooling across channels:

$$F_{\text{avg}} = \text{Mean}_c(F),$$

and maximum pooling across channels:

$$F_{\max} = \text{Max}_c(F).$$

Both tensors have dimensions

$$H \times W \times 1.$$

Concatenate them:

$$F_s = [F_{\text{avg}} \| F_{\max}].$$

The spatial attention map is then computed as

$$M_s(F) = \sigma\left(\text{Conv}_{7\times 7}(F_s)\right).$$

Therefore,

$$M_s(F) \in \mathbb{R}^{H\times W\times 1}.$$

The spatially refined feature is

$$F'_s = F \odot M_s(F).$$

Spatial attention answers:

$$\boxed{\text{Which spatial locations are important?}}$$

11. Sequential Channel–Spatial Attention

Channel and spatial attention can be sequentially applied.
First,

$$F_c = M_c(F) \odot F.$$

Then,

$$F_{cs} = M_s(F_c) \odot F_c.$$

The overall structure becomes:

$$\boxed{F \to \text{Channel Attention} \to \text{Spatial Attention} \to F_{cs}}$$

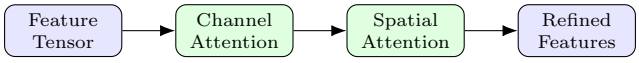
This arrangement first determines

"what is important"

and then determines

"where it is important."

A simplified diagram is shown below.



12. Self-Attention

Self-attention allows spatial locations to interact directly with other spatial locations.
For

$$F \in \mathbb{R}^{H \times W \times C},$$

flatten the spatial dimensions:

$$X \in \mathbb{R}^{N \times C},$$

where

$$N = HW.$$

Generate the query, key, and value matrices:

$$Q = XW_Q,$$

$$K = XW_K,$$

and

$$V = XW_V.$$

The attention matrix is

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right),$$

where d is the dimension of the query/key representation.
The attended output is

$$Y = AV.$$

After reshaping the output back to the original spatial structure, a residual formulation may be used:

$$F_{\text{SA}} = F + \gamma\,\text{reshape}(Y),$$

where γ is a learnable scaling parameter.
Self-attention attempts to answer:

Which spatial features are related to other spatial features?

Since the attention matrix has dimension approximately

$$N \times N,$$

its computational requirements increase rapidly as

$$N = HW$$

increases.
Therefore, self-attention should preferably be introduced at

$$P_3$$

or

$$P_4.$$

13. Inserting Attention into ResNet50

For any selected stage P_k, the network structure can be written as

$$\text{Previous ResNet Stage} \rightarrow \text{Attention} \rightarrow \text{Next ResNet Stage}.$$

For example, attention inserted after conv4_x gives:

$$\boxed{\text{conv4_x} \rightarrow \text{Attention} \rightarrow \text{conv5_x}}$$

For channel–spatial attention:

$$\boxed{\text{conv4_x} \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow \text{conv5_x}}$$

The student must record the feature dimensions immediately before and after the attention block.
Attention should not unintentionally modify the spatial or channel dimensions unless the selected mechanism explicitly requires such a transformation.

14. Main Experiment 1: Attention-Type Ablation

The first experiment studies the effect of different types of attention.
Fix the attention position at

$$\boxed{P_3 = \text{after conv4_x}}$$

for all models.
Compare the following:

Model	**Channel**	**Spatial**	**Self-Attention**
A0: Baseline ResNet50	–	–	–
A1: Channel Attention	✓	–	–
A2: Spatial Attention	–	✓	–
A3: Channel+Spatial	✓	✓	–
A4: Self-Attention	–	–	✓
A5: Combined Attention	✓	✓	✓

This experiment investigates:

$$\boxed{\text{How does attention type affect model performance?}}$$

Students must keep the following identical:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;
- learning rate schedule;
- epochs;
- batch size;
- backbone;
- classifier;
- attention position; and
- evaluation procedure.

15. Main Experiment 2: Attention-Position Ablation

The second experiment investigates the influence of the position of attention.
Fix the attention type as:

$$\boxed{\text{Channel Attention} \rightarrow \text{Spatial Attention}}$$

and insert exactly the same module at different stages.

Experiment	Attention Position	Resolution	Channels
B1	After conv2_x	56×56	256
B2	After conv3_x	28×28	512
B3	After conv4_x	14×14	1024
B4	After conv5_x	7×7	2048

Students should study whether attention is more beneficial for:

- high-resolution low-level features;
- intermediate representations; or
- low-resolution semantic features.

The principal question is:

$$\boxed{\text{How does the location of attention influence performance?}}$$

16. Main Experiment 3: Multiple Attention Positions

Attention may also be inserted at more than one ResNet50 stage.
Compare:

$$\text{Attention only at } P_3,$$

$$\text{Attention only at } P_4,$$

and

$$\text{Attention at } P_3 + P_4.$$

As an extended experiment, students may evaluate:

$$P_2 + P_3 + P_4.$$

The purpose is to investigate whether applying attention repeatedly provides sufficient performance improvement to justify the additional:

- parameters;
- computation;
- memory;
- training time; and
- inference time.

Students should experimentally answer:

$$\boxed{\text{Is attention at more locations necessarily more effective?}}$$

17. Main Experiment 4: Attention Order

The sequence in which channel and spatial attention are applied may influence the resulting features.
Compare:

$$\boxed{\text{Channel} \rightarrow \text{Spatial}}$$

against

$$\boxed{\text{Spatial} \rightarrow \text{Channel}}$$

while maintaining the same insertion position.
For example, at P_3:

$$F \rightarrow M_c \rightarrow M_s$$

should be compared with

$$F \rightarrow M_s \rightarrow M_c.$$

Students should determine whether selecting informative feature channels before selecting spatial regions produces behavior different from selecting spatial regions first.

18. Main Experiment 5: Spatial Attention Kernel Size

The receptive field used to generate the spatial attention map may influence the result.
Evaluate:

$$k \in \{3, 5, 7\}.$$

Use the same attention position and training protocol.

Spatial Kernel	Accuracy (%)	Macro F1 (%)	Inference Time (ms/image)
3×3	92.18	92.28	16.86
5×5	92.73	92.77	15.92
7×7	91.82	92.00	16.00

Students should discuss whether a larger spatial kernel provides a useful increase in contextual information.

19. Additional Task: Comparison of Different Attention Mechanisms

As an additional exploration task, students should compare the attention mechanisms implemented in the laboratory with established attention approaches.
Students should select at least

$$\boxed{\text{two additional attention mechanisms}}$$

from the following:

- Squeeze-and-Excitation (SE) Attention;
- Efficient Channel Attention (ECA);
- Convolutional Block Attention Module (CBAM);
- Coordinate Attention; and
- lightweight Self-Attention.

The pretrained backbone must remain:

$$\boxed{\text{ResNet50}}$$

for all comparisons.
To ensure a fair comparison, additional attention mechanisms should preferably be inserted at the same position:

$$\boxed{P_3 = \text{after conv4_x}}$$

unless the student provides a technical justification for using another location.

19.1 Suggested Comparison

Attention	Accuracy (%)	Precision (%)	Recall (%)	Macro F1 (%)	Parameters	Inference Time (ms/image)
No Attention	93.27	93.41	93.29	93.34	23,597,957	18.60
Channel / SE	92.91	92.93	92.97	92.93	23,730,117	17.03
Spatial Attention	91.82	92.21	91.83	92.00	23,598,055	16.00
ECA	92.00	92.47	92.03	92.18	23,597,962	17.10
CBAM	92.00	92.47	92.01	92.14	23,730,215	16.24
Self-Attention	94.73	94.64	94.92	94.74	24,122,246	17.56
Additional Mechanism	93.45	93.70	93.51	93.56	24,254,504	17.05

19.2 Questions for Attention-Mechanism Comparison

Students should investigate:

1. How does classification accuracy vary across attention mechanisms?
2. How does macro F1-score vary across attention mechanisms?
3. How many parameters are introduced by each attention module?
4. How does each attention mechanism influence inference time?
5. Does attention improve the convergence behavior?
6. Does attention produce more focused feature localization?
7. Does an increase in attention complexity always improve classification performance?
8. Does a lightweight attention mechanism provide comparable performance with a computationally heavier mechanism?

The objective is to understand:

$$\boxed{\text{Performance} + \text{Complexity} + \text{Attention Behaviour}}$$

rather than considering only classification accuracy.

20. Transfer-Learning Strategy

Use ImageNet pretrained weights.

A recommended training strategy is:

1. Load ResNet50 with ImageNet pretrained weights.
2. Remove the original ImageNet classification layer.
3. Insert the selected attention module.
4. Add Global Average Pooling.
5. Add Dropout.
6. Add a Dense classification layer containing five output neurons.
7. Initially freeze the pretrained ResNet50 layers.
8. Train the newly introduced attention and classification layers.
9. Unfreeze the final ResNet50 stage or selected final stages.
10. Fine-tune the network using a smaller learning rate.

A suitable initial learning rate for the newly introduced layers is

$$10^{-3}.$$

During fine-tuning, a smaller learning rate such as

$$10^{-5} \quad \text{to} \quad 10^{-4}$$

may be used.

The exact values selected by the student should remain identical for all controlled comparisons.

21. Loss Function

For the five-class classification problem, use categorical cross-entropy.

For N samples:

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{5} y_{i,c}\log \widehat{y}_{i,c},$$

where:

$$y_{i,c}$$

is the ground-truth label and

$$\widehat{y}_{i,c}$$

is the predicted probability.

The same loss formulation must be used for every attention comparison.

22. Evaluation Metrics

Students must report the following quantitative metrics:

- classification accuracy;
- macro precision;
- macro recall;
- macro F1-score;
- number of trainable parameters;
- total number of parameters; and
- average inference time per image.

For class c,

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c},$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c},$$

and

$$F1_c = 2\frac{\text{Precision}_c \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}.$$

Macro F1 is:

$$\text{Macro F1} = \frac{1}{C}\sum_{c=1}^{C} F1_c,$$

where

$$C = 5.$$

23. Required Main Result Table

Students should summarize the major experiments using the following table.

Model	**Position**	**Accuracy (%)**	**Precision (%)**	**Recall (%)**	**F1 (%)**	**Parameters**	**Inference Time (ms/image)**
Baseline	–	93.27	93.41	93.29	93.34	23,597,957	18.60
Channel Attention	P_3	92.73	92.92	92.96	92.92	23,730,117	17.32
Spatial Attention	P_3	91.82	92.21	91.83	92.00	23,598,055	16.00
Channel+Spatial	P_2	91.45	91.61	91.52	91.55	23,631,367	16.99
Channel+Spatial	P_3	90.36	90.44	90.45	90.42	23,730,215	17.66
Channel+Spatial	P_4	92.55	92.68	92.56	92.58	24,124,519	17.40
Self-Attention	P_3	94.73	94.64	94.92	94.74	24,122,246	17.56
Additional Attention-1 (SE)	P_3	92.91	92.93	92.97	92.93	23,730,117	17.03
Additional Attention-2 (ECA)	P_3	92.00	92.47	92.03	92.18	23,597,962	17.10

24. Graphical Performance Analysis

Students should not present only numerical tables.
Appropriate graphs must be generated to analyze model behavior.
At least three of the following graphical comparisons should be included in the laboratory report.

24.1 Training and Validation Accuracy

Plot

$$\text{Epoch} \quad \text{versus} \quad \text{Accuracy}$$

for:

- baseline ResNet50; and
- selected attention models.

Both training and validation accuracy should be shown.
Students should comment on whether attention:

- accelerates convergence;
- produces smoother learning;
- reduces the train–validation gap;
- leads to overfitting; or
- causes unstable training.

24.2 Training and Validation Loss

Plot

$$\text{Epoch} \quad \text{versus} \quad \mathcal{L}_{CE}.$$

The graph should include training and validation loss.
Students should compare the convergence behavior of the baseline and attention models.

24.3 Attention Mechanism versus Accuracy

Generate a bar graph:

$$x = \text{Attention Mechanism}$$

and

$$y = \text{Classification Accuracy}.$$

Possible entries are:

$$\text{Baseline, Channel, Spatial, Channel+Spatial, Self-Attention, SE, ECA, CBAM.}$$

24.4 Attention Mechanism versus Macro F1

Similarly, plot:

$$x = \text{Attention Mechanism}$$

and

$$y = \text{Macro F1-score}.$$

This comparison is useful when the class-wise performance is not uniform.

24.5 Accuracy versus Model Parameters

Create a scatter plot using:

$$x = \text{Total Model Parameters}$$

and

$$y = \text{Test Accuracy}.$$

Each point should represent one attention mechanism.
This allows the student to study the

$$\boxed{\text{Accuracy–Complexity Trade-off}}$$

instead of evaluating accuracy independently.

24.6 Macro F1 versus Inference Time

Plot:

$$x = \text{Inference Time per Image}$$

and

$$y = \text{Macro F1-score.}$$

Students should analyze whether additional computational cost produces a meaningful increase in classification performance.

25. Confusion Matrix Analysis

Generate a confusion matrix for:

1. baseline ResNet50; and
2. the selected attention-based ResNet50.

The confusion matrix should contain all five flower classes.

Students should analyze:

- which classes are most accurately classified;
- which classes are frequently confused;
- which class benefits most from attention;
- whether attention reduces a particular type of confusion; and
- whether the improved overall accuracy is distributed across classes.

26. Qualitative Analysis of Attention

Quantitative classification performance does not directly explain where the network obtains evidence for its prediction.

Students must therefore perform qualitative analysis.

The visual analysis should include:

- internal attention maps;
- attention overlays;
- intermediate feature maps; and
- Grad-CAM explanations.

Students should preferably use the same input image when comparing different models.

27. Grad-CAM Visualization

Gradient-weighted Class Activation Mapping (Grad-CAM) provides a class-specific visualization of the regions contributing to a prediction.

Let

$$A^k$$

represent feature map k from a selected convolutional layer.

For class score y^c, the importance coefficient is:

$$\alpha_k^c = \frac{1}{HW} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}.$$

The Grad-CAM map is:

$$L_{\text{Grad}-\text{CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k \right).$$

The map should be resized to the original image resolution and overlaid on the input.

Students should compare:

Input | Baseline Grad-CAM | Attention Model Grad-CAM

for the same image.

27.1 Grad-CAM Interpretation

Students should qualitatively determine whether the attention model:

- focuses more strongly on the flower;
- suppresses irrelevant background regions;
- concentrates on petals;
- concentrates on the central flower structure;
- attends to discriminative edges;
- distributes activation across the entire flower;
- incorrectly focuses on leaves;
- incorrectly focuses on background objects; or
- changes its focus for an incorrect prediction.

Students should avoid unsupported statements such as:

> "The Grad-CAM of Model A is better."

Instead, the observation should describe visible evidence.
For example:

> The baseline network activates both the flower and surrounding leaves, whereas the attention-based model produces stronger activation over the petal region and weaker activation over the background.

28. Internal Attention Maps

An important objective of this experiment is to distinguish between:

Internal Attention

and

Post-hoc Explanation

because they are not the same.
An attention module directly computes weights during the forward pass.
Grad-CAM, in contrast, is generated after prediction using gradients.

28.1 Channel-Attention Visualization

For channel attention:

$$M_c(F) = [m_1, m_2, \ldots, m_C].$$

Students should visualize the learned channel importance values using a bar graph.
Instead of displaying every channel when C is large, the student may show the top-K channels.
Recommended values are:

$$K = 10$$

or

$$K = 20.$$

Students should comment on whether:

- a small number of channels receive strong weights;
- many channels contribute similarly; or
- channel weighting changes substantially for different input images.

28.2 Spatial-Attention Visualization

For spatial attention:

$$M_s(F) \in \mathbb{R}^{H \times W}.$$

Resize the attention map to:

$$224 \times 224$$

and overlay it on the input image.
Display:

$$\boxed{\text{Input} \mid \text{Spatial Attention Map} \mid \text{Attention Overlay}}$$

Students should identify the image regions receiving high attention values.

28.3 Self-Attention Visualization

For self-attention, students may visualize one of the following:

- attention associated with a selected query location;
- mean attention across query positions;
- maximum attention across query positions; or
- an aggregated spatial self-attention response.

The visualization should be reshaped into a spatial form and, when appropriate, resized to the image resolution.
Students should analyze how distant regions interact within the feature map.

29. Attention Position Visualization

Use the same test image to visualize spatial attention at different ResNet50 stages.
Generate:

$$M_s^{P_2}, \qquad M_s^{P_3}, \qquad M_s^{P_4}.$$

Display the maps as:

$$\boxed{\text{Input} \mid M_s^{P_2} \mid M_s^{P_3} \mid M_s^{P_4}}$$

The objective is to visually study how attention changes as feature representations become deeper.
Possible observations may include:

- P_2 responding strongly to edges, colors, textures, and fine structures;
- P_3 responding to flower parts and intermediate object structures;
- P_4 responding to larger semantic flower regions.

These observations must be based on the student's experimental visualizations.

30. Qualitative Comparison Across Attention Mechanisms

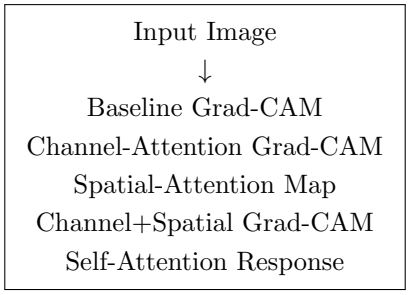
Select at least:

$$\boxed{\text{3 correctly classified images}}$$

and

$$\boxed{\text{2 incorrectly classified images.}}$$

For every selected image, compare the behavior of multiple models.
A recommended arrangement is:



Students may alternatively arrange the images horizontally in the report.

30.1 Qualitative Criteria

The student should discuss:

- foreground localization;
- background suppression;
- localization of petals;
- localization of the flower center;
- activation compactness;
- activation coverage;
- incorrectly attended regions;
- differences between correctly and incorrectly classified samples;
- differences between shallow and deep attention; and
- consistency of attention across different images.

31. Required Visualizations

The laboratory report must contain the following visualizations:

1. sample input images from all five classes;
2. training and validation accuracy curves;
3. training and validation loss curves;
4. confusion matrix;
5. feature map before attention;
6. channel-attention weights;
7. spatial-attention heatmap;
8. spatial-attention overlay;
9. self-attention response for the self-attention experiment;
10. feature map after attention;
11. baseline Grad-CAM;
12. Grad-CAM for at least one attention-based model;
13. attention maps at P_2, P_3, and P_4; and
14. at least one graphical comparison of the performances of different attention mechanisms.

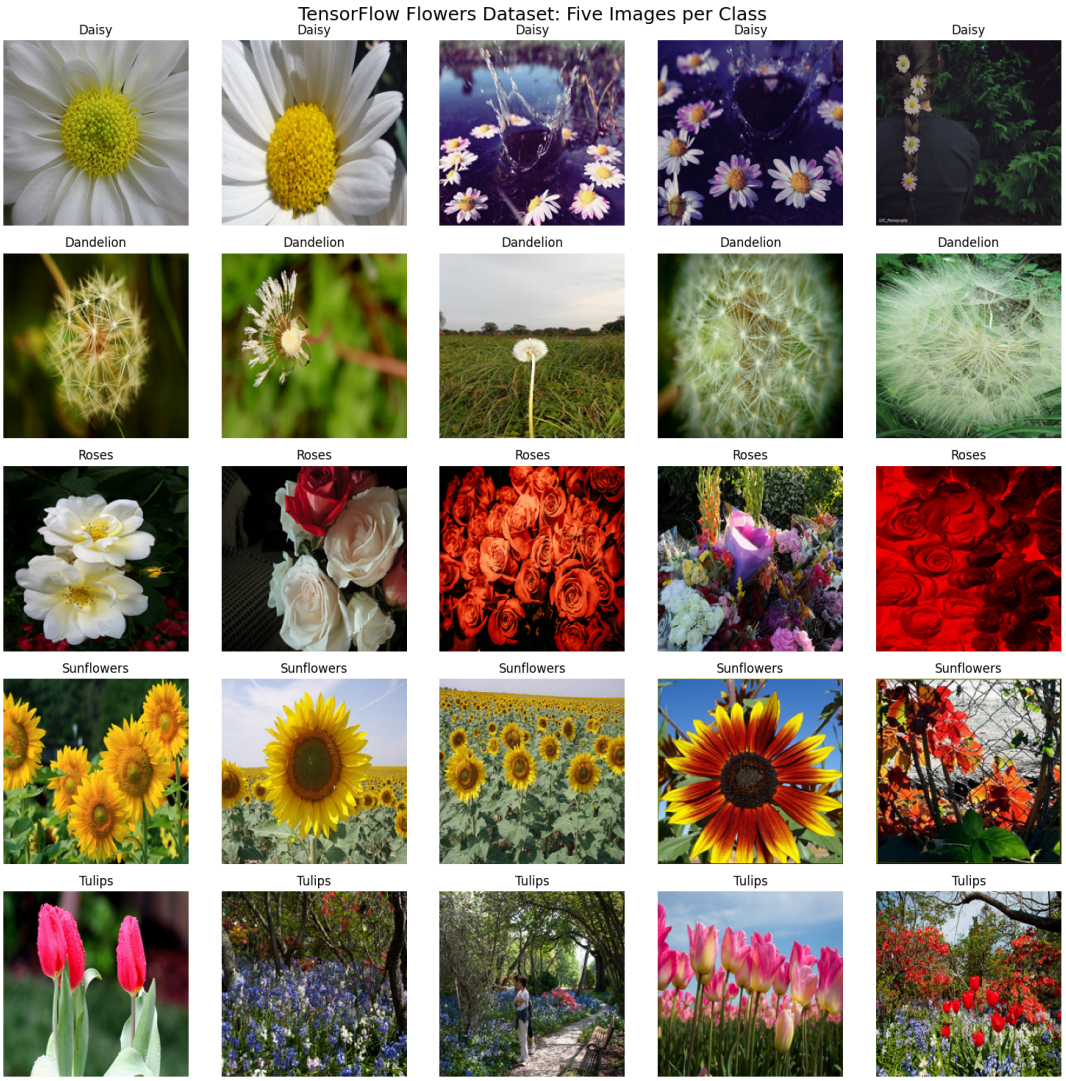


Figure 1: Sample visualization from the experiment

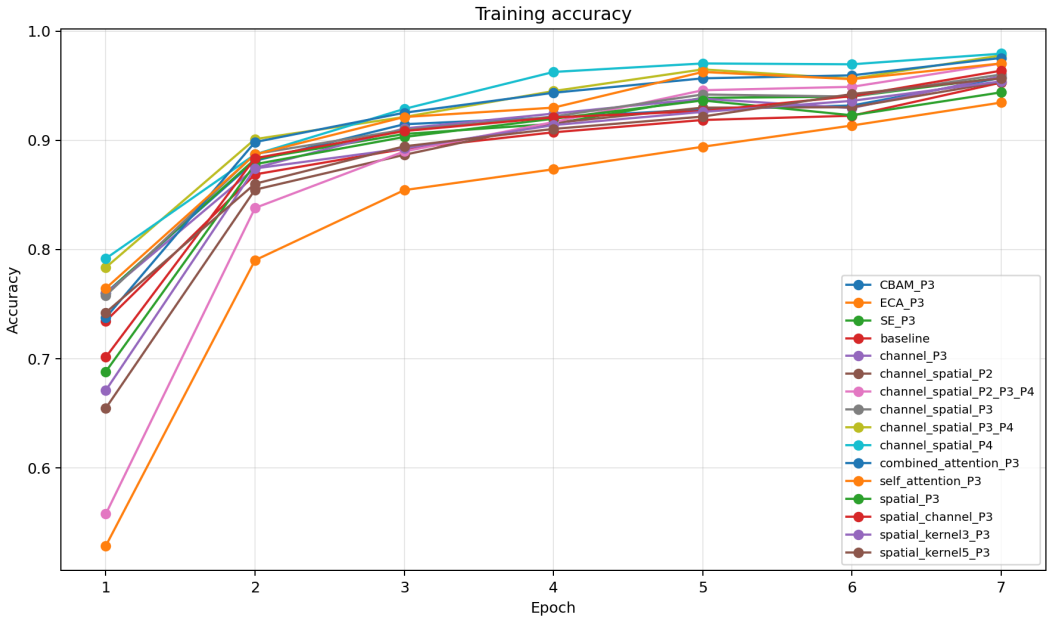


Figure 2: Sample visualization from the experiment

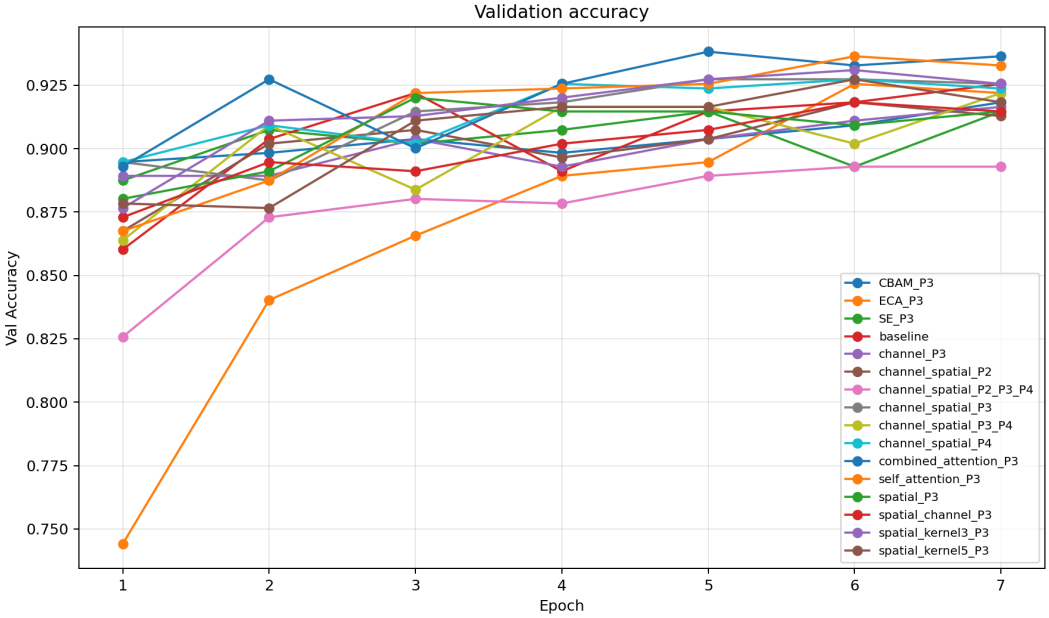


Figure 3: Sample visualization from the experiment

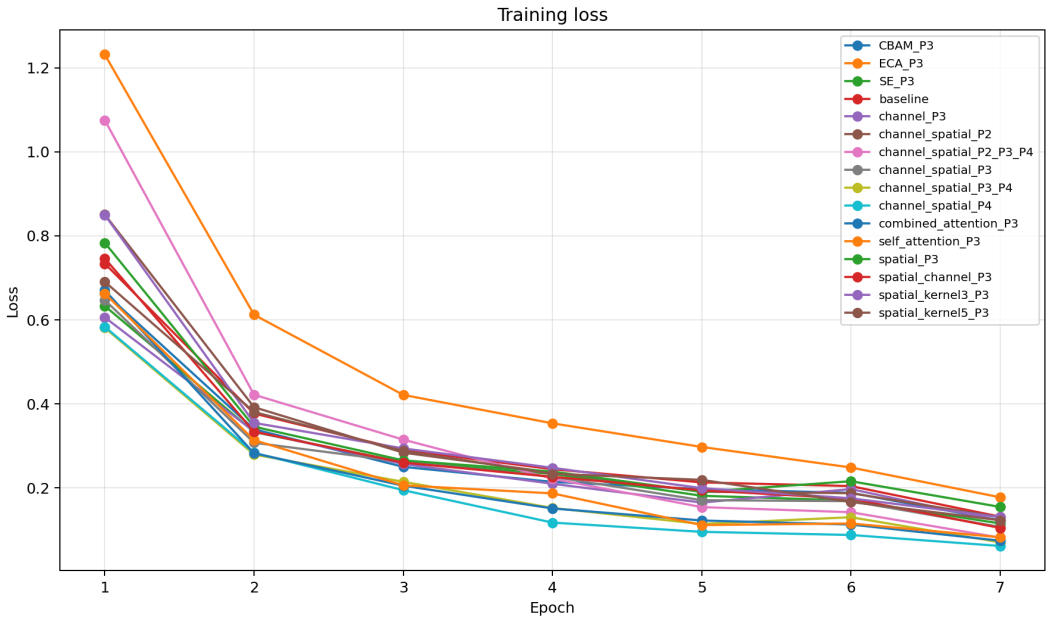


Figure 4: Sample visualization from the experiment

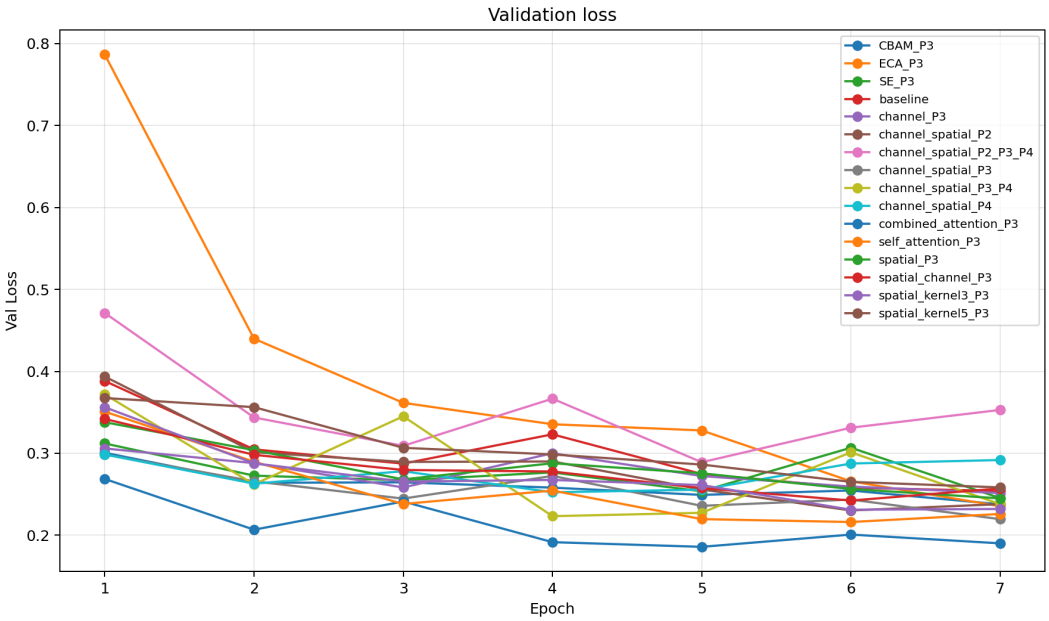


Figure 5: Sample visualization from the experiment

25. Confusion Matrix Analysis

25.1 Baseline Model

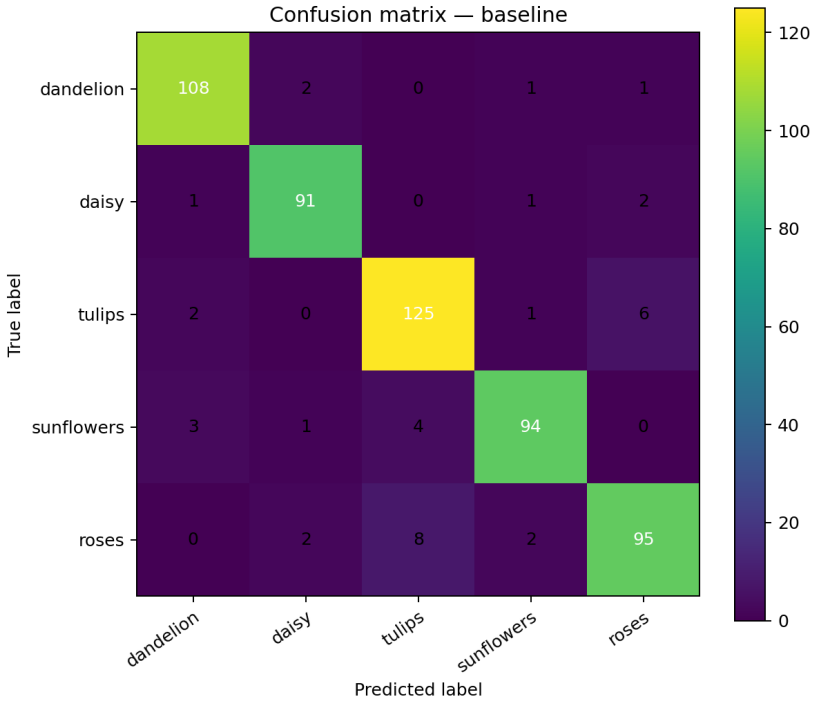


Figure 6: Confusion matrix of baseline ResNet50

25.2 Attention-Type Ablation

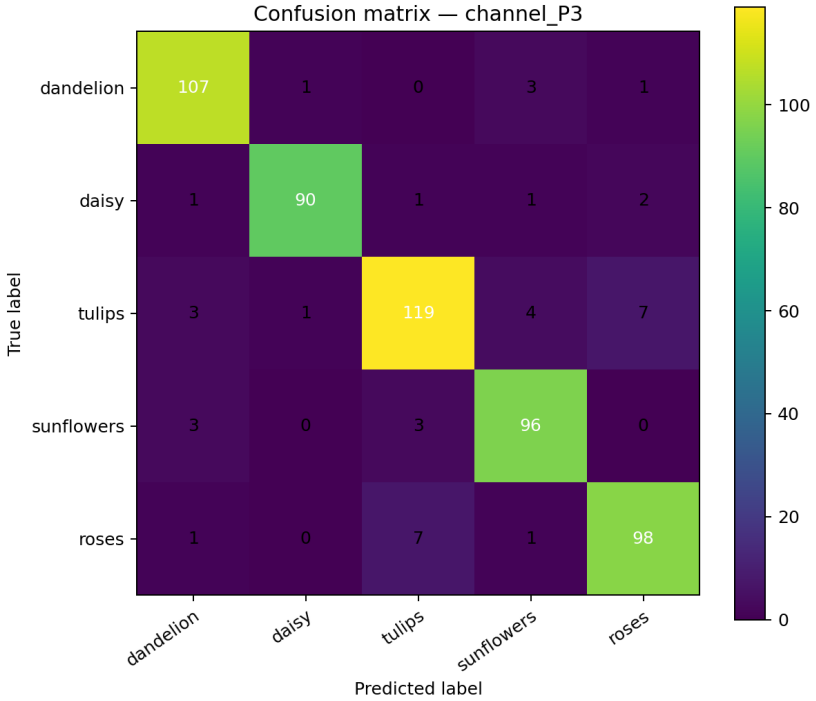


Figure 7: Confusion matrix with channel attention at P3

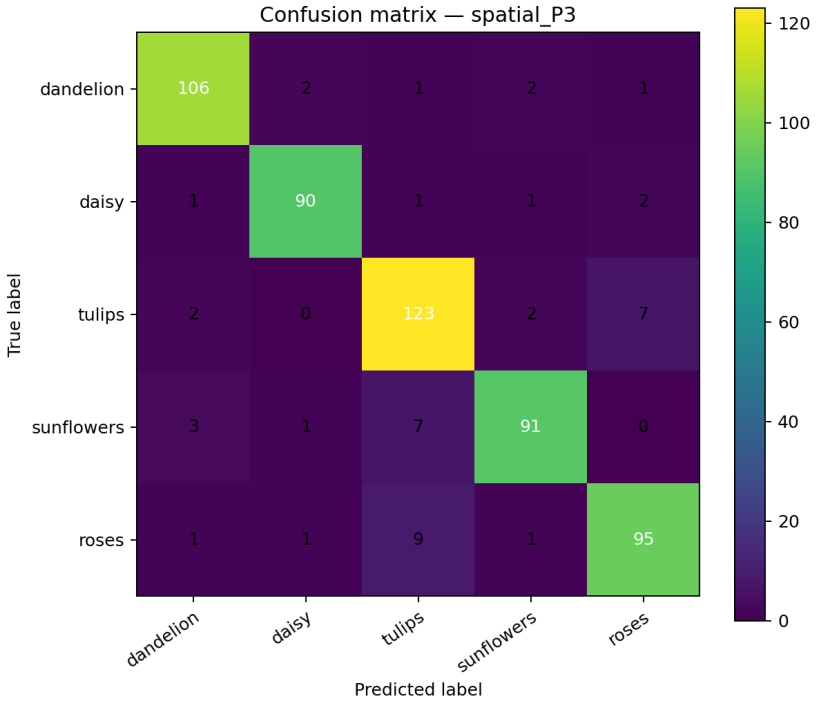


Figure 8: Confusion matrix with spatial attention at P3

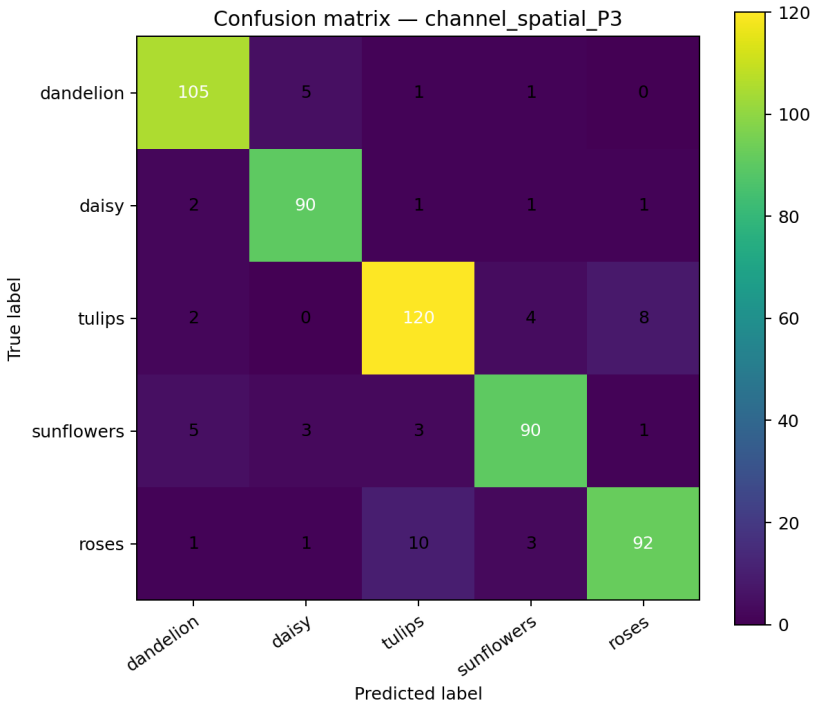


Figure 9: Confusion matrix with channel-spatial attention at P3

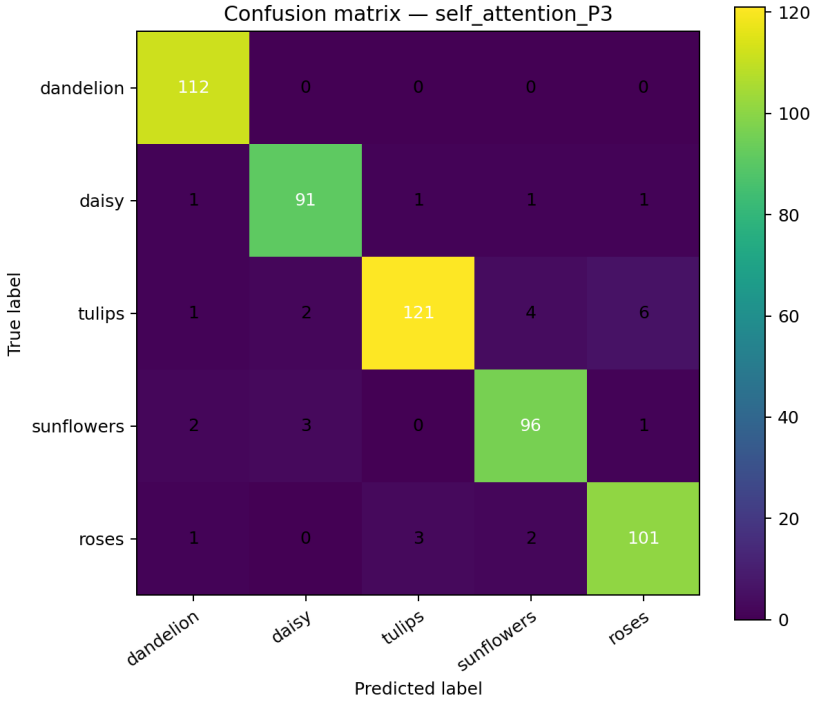


Figure 10: Confusion matrix with self-attention at P3

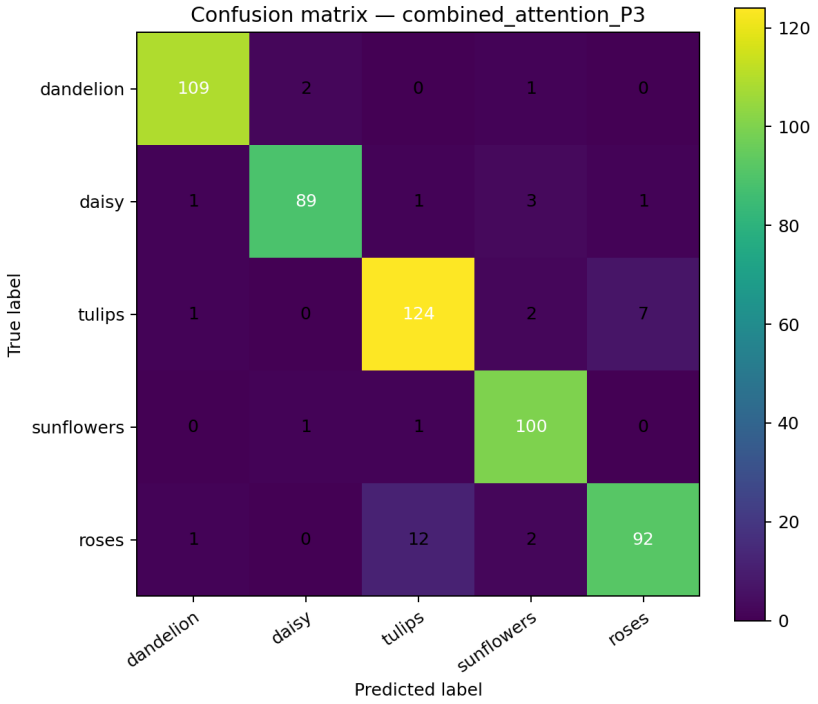


Figure 11: Confusion matrix with combined attention at P3

Attention-Position Ablation

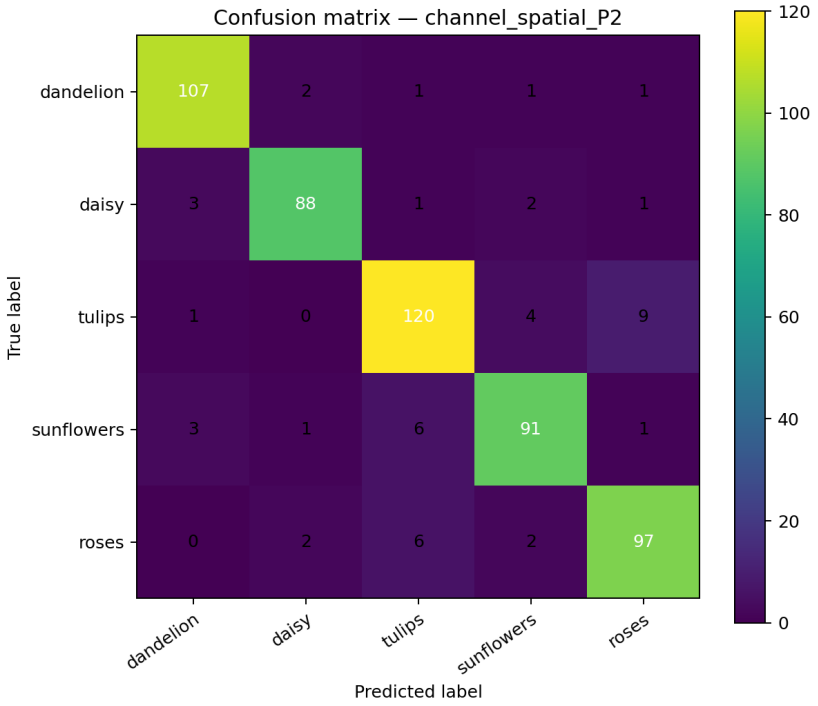


Figure 12: Channel-spatial attention at P2

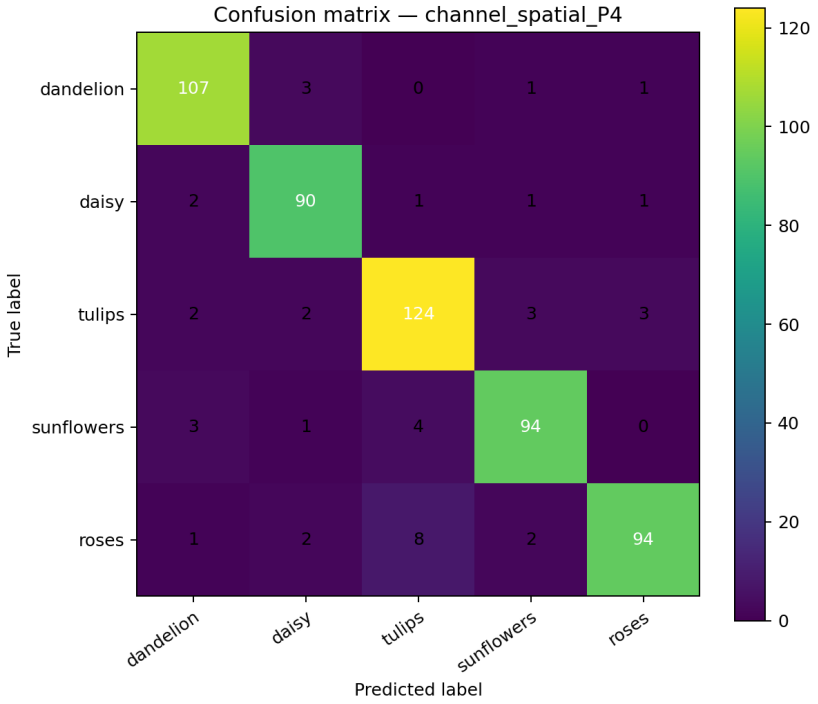


Figure 13: Channel-spatial attention at P4

Multiple Attention Positions

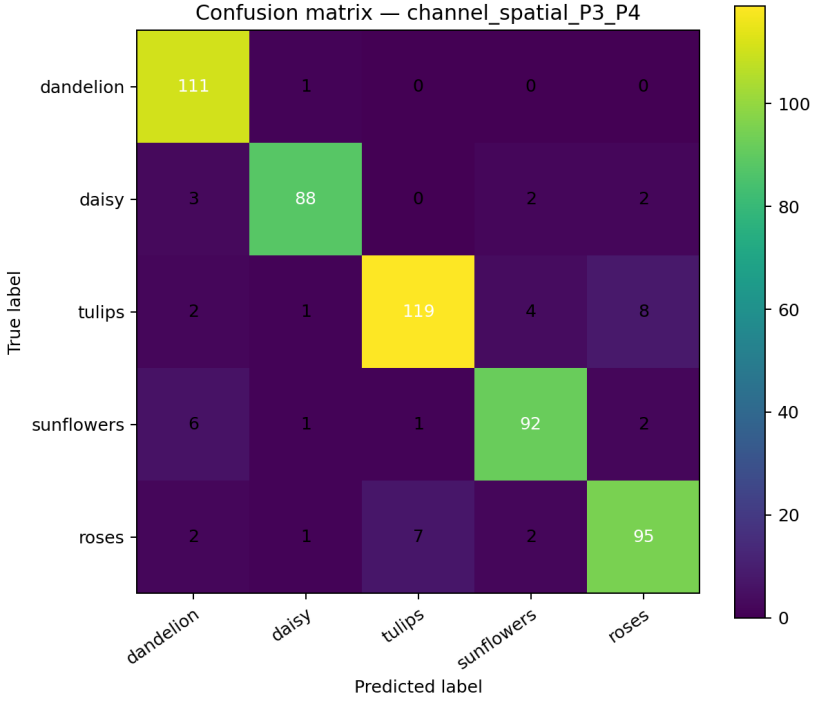


Figure 14: Channel-spatial attention at P3 and P4

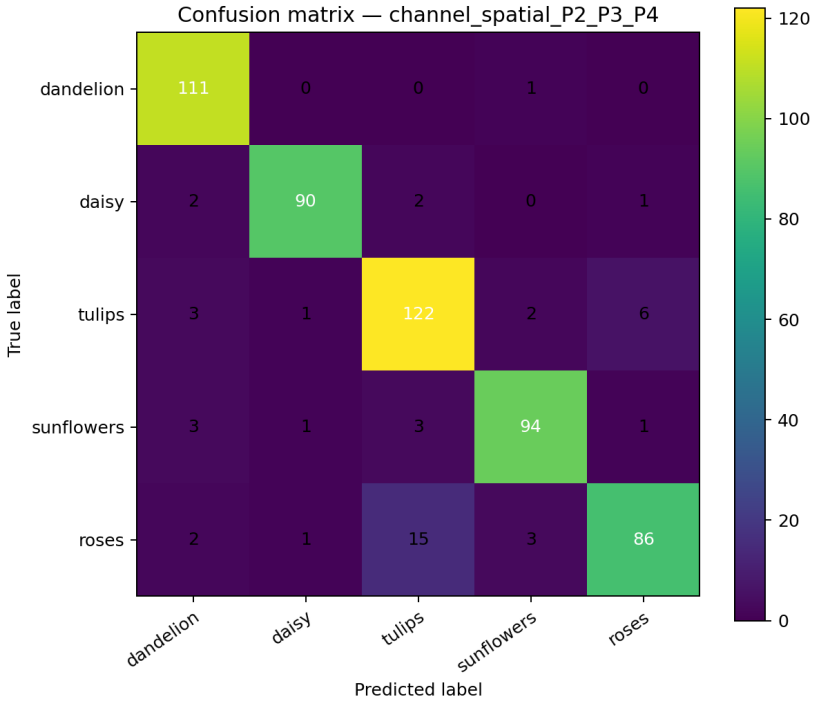


Figure 15: Channel-spatial attention at P2, P3 and P4

Attention-Order Comparison

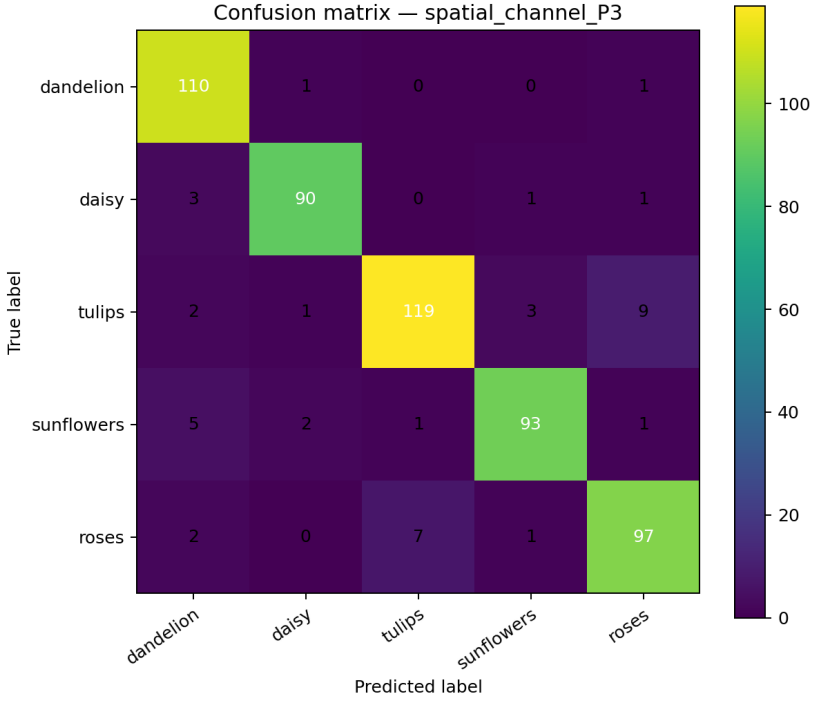


Figure 16: Spatial attention followed by channel attention at P3

Spatial Attention Kernel Comparison

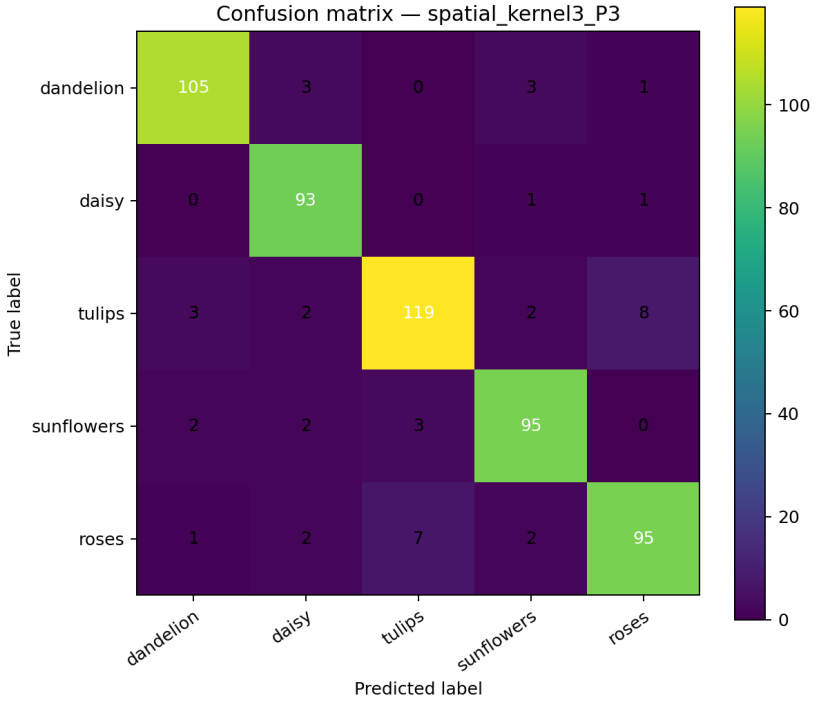


Figure 17: Spatial attention with a 3 x 3 kernel

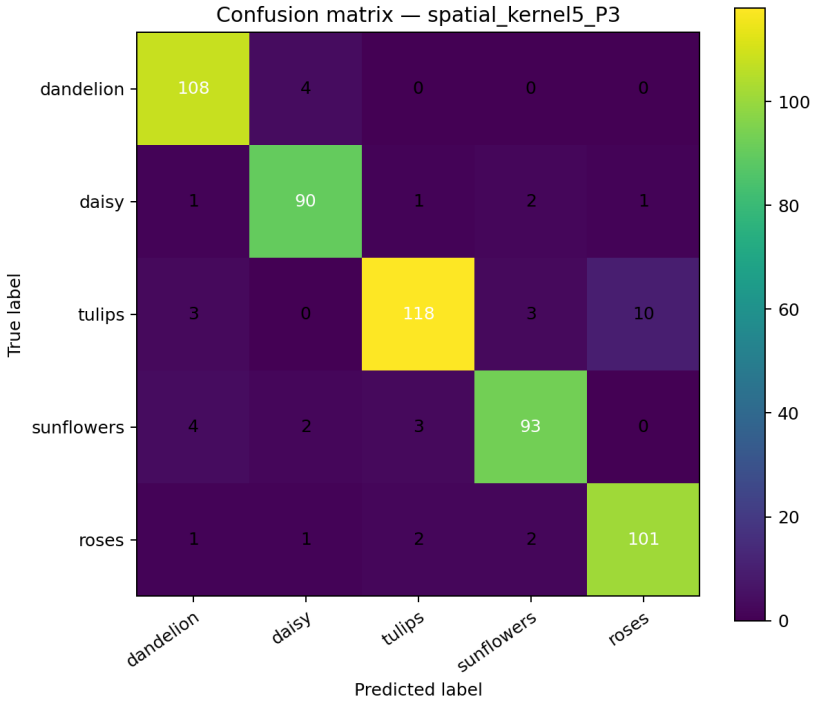


Figure 18: Spatial attention with a 5 x 5 kernel

Additional Attention Mechanisms

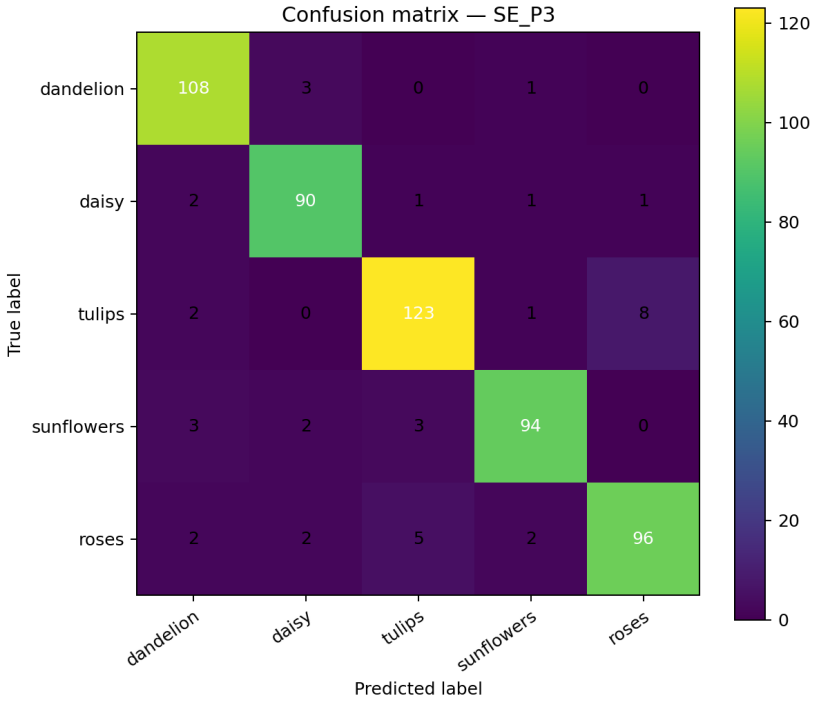


Figure 19: Squeeze-and-Excitation attention at P3

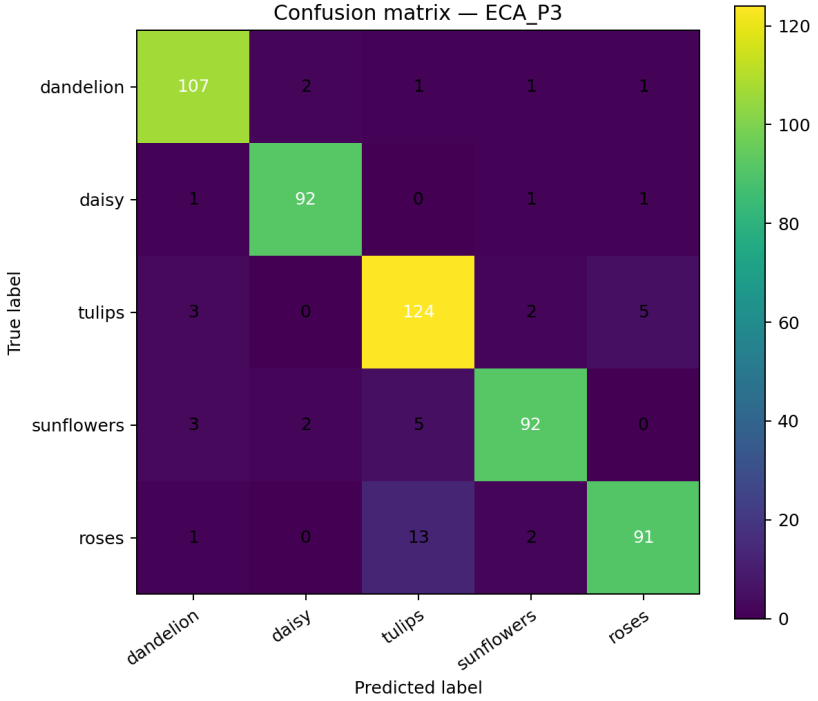


Figure 20: Efficient Channel Attention at P3

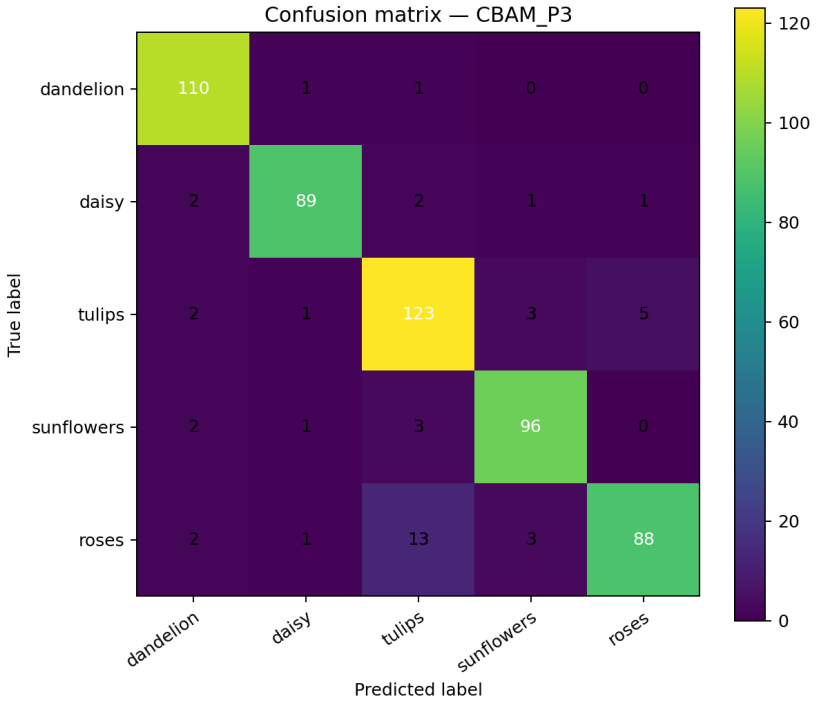


Figure 21: CBAM attention at P3

5. Feature Map Before Attention

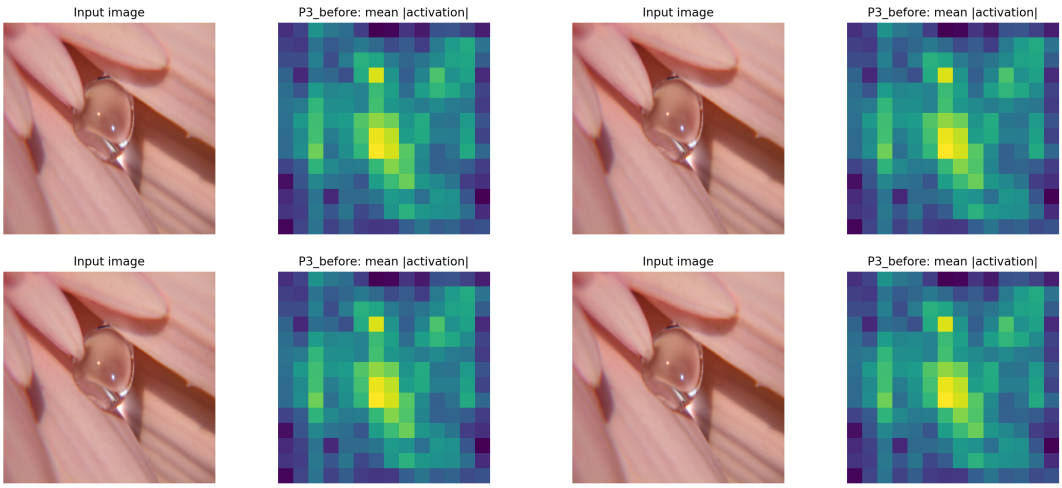


Figure 22: Feature maps before applying different attention mechanisms.

6. Channel-Attention Weights

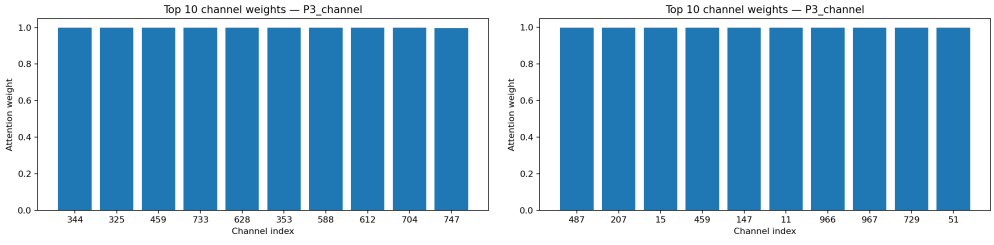


Figure 23: Comparison of channel-attention weights for the channel and channel-spatial attention mechanisms.

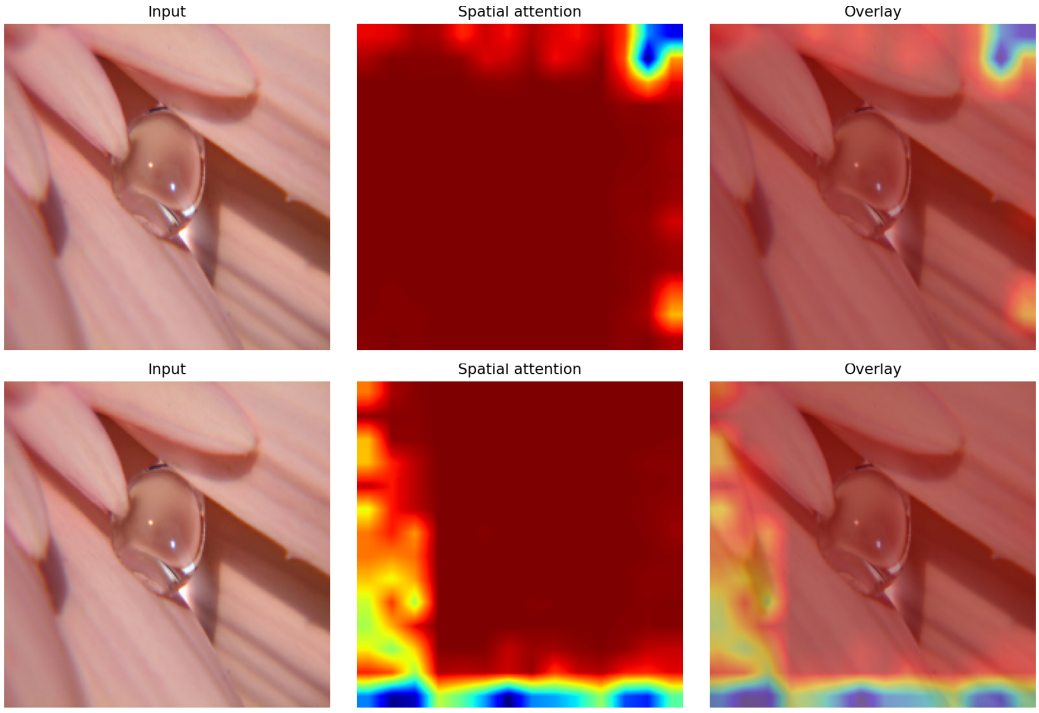


Figure 24: Spatial-attention heatmap visualizations.

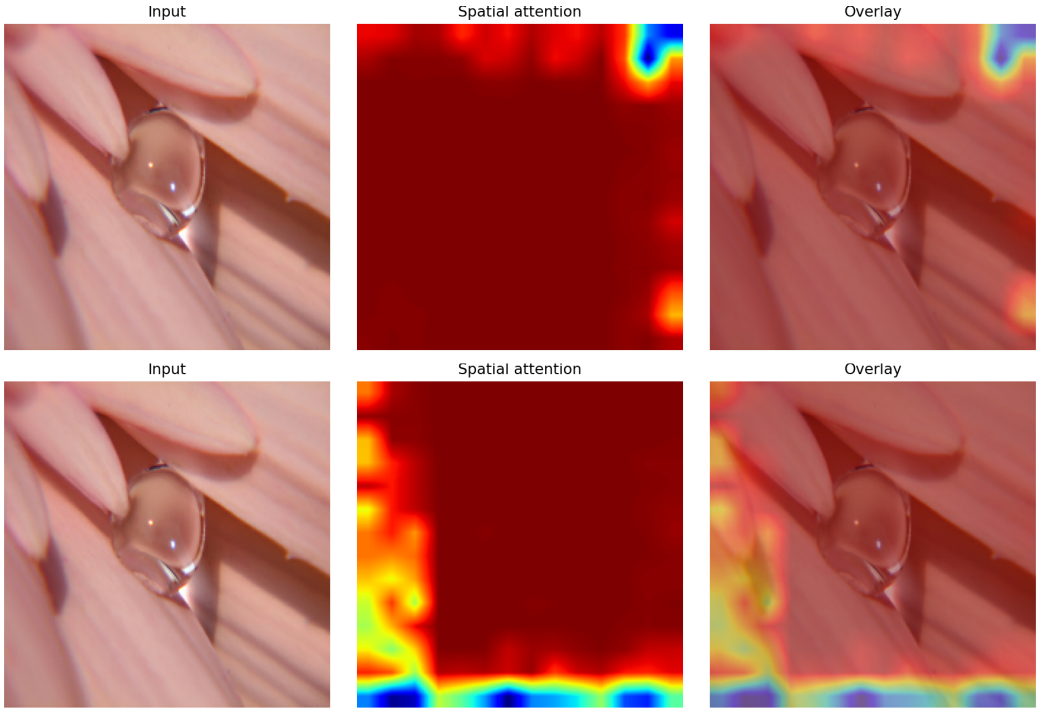


Figure 25: Spatial-attention overlays highlighting important image regions.

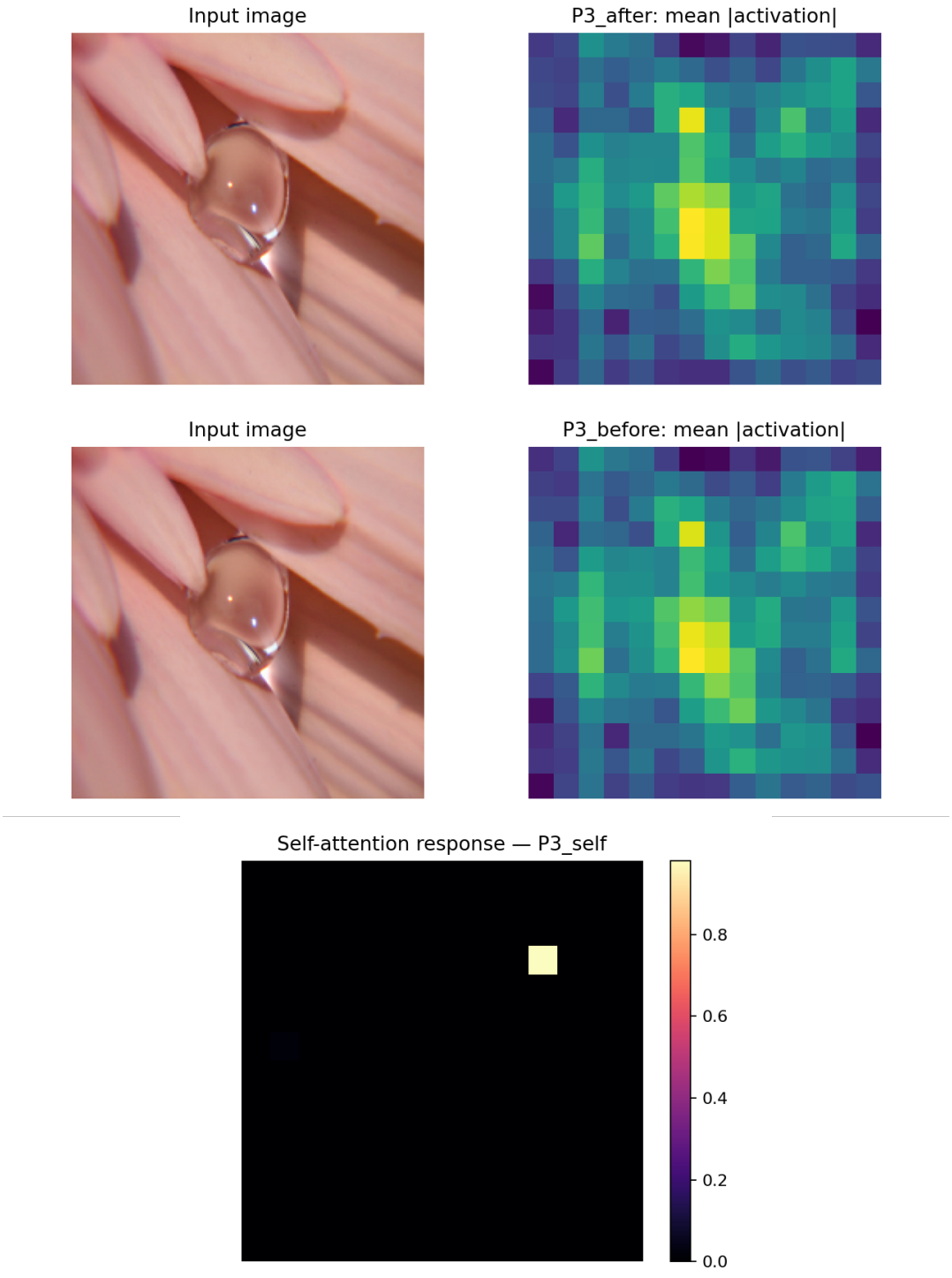


Figure 26: Self-attention response and associated feature maps.

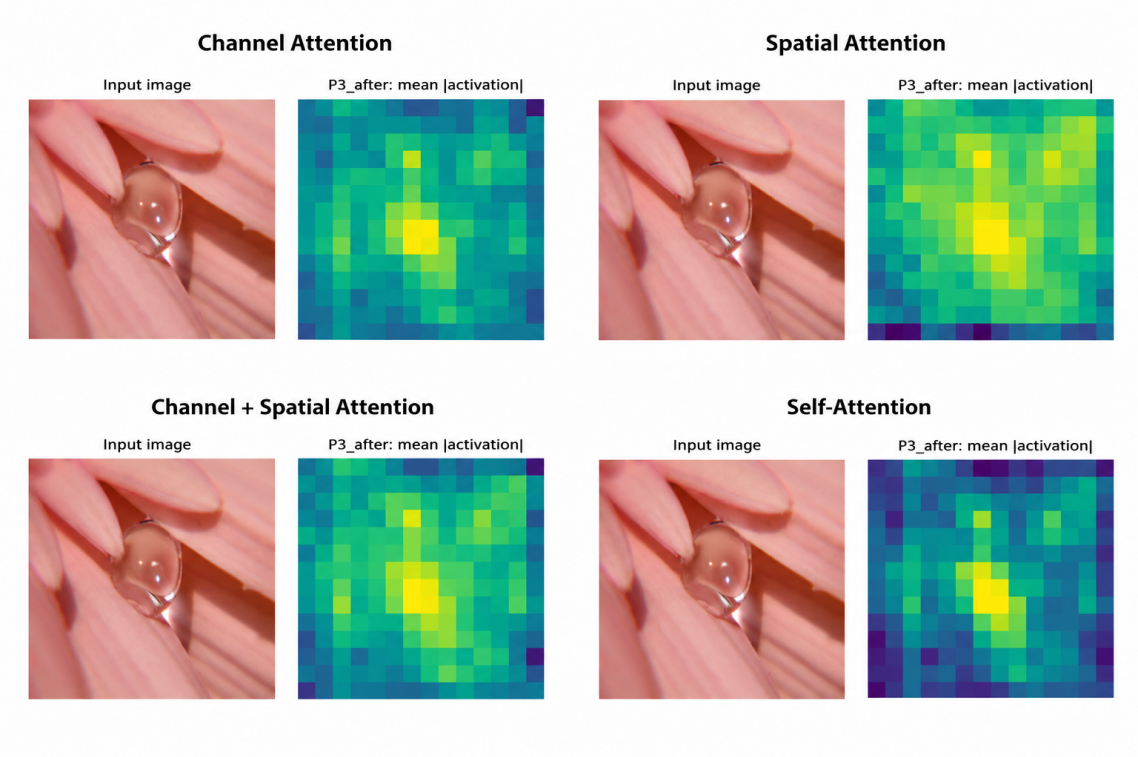


Figure 27: Feature maps after applying the attention mechanisms.

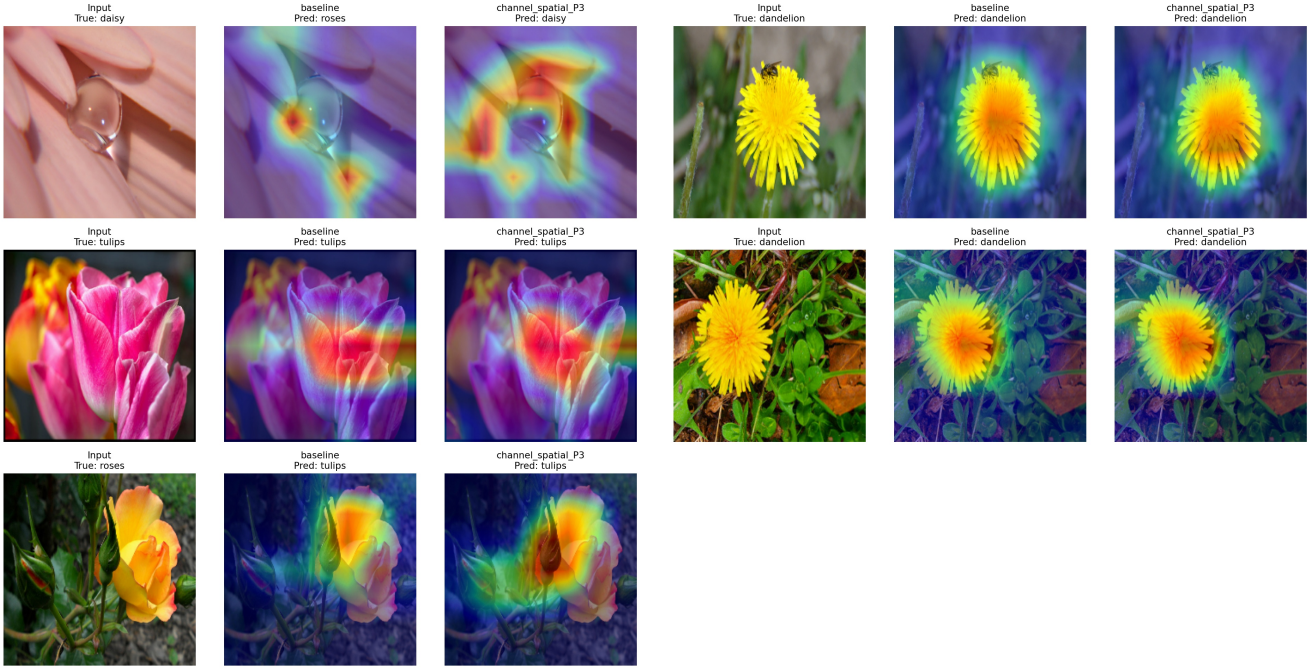


Figure 28: Grad-CAM visualizations for the baseline model.

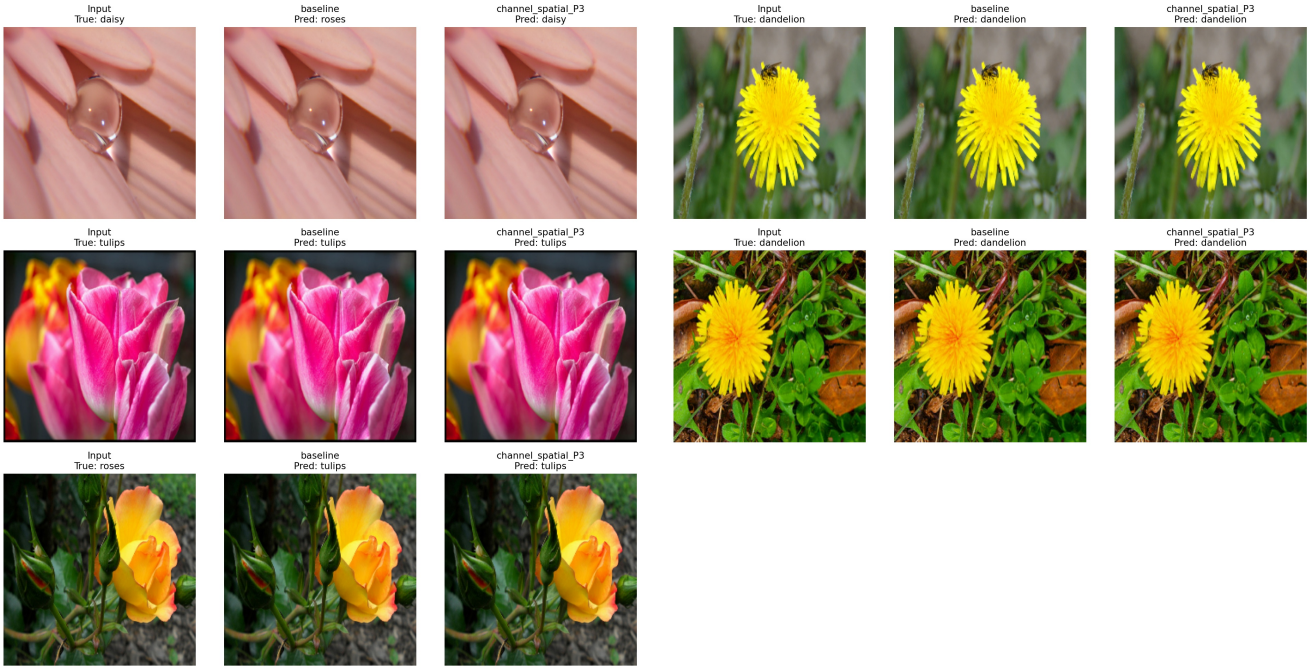


Figure 29: Grad-CAM visualizations for an attention-based model.

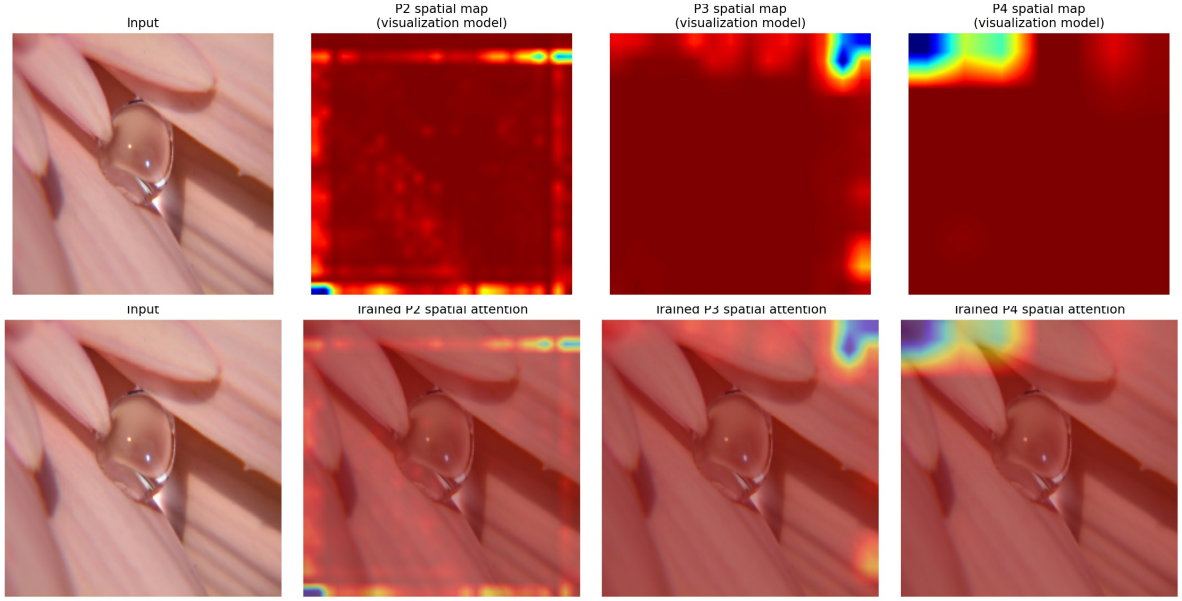
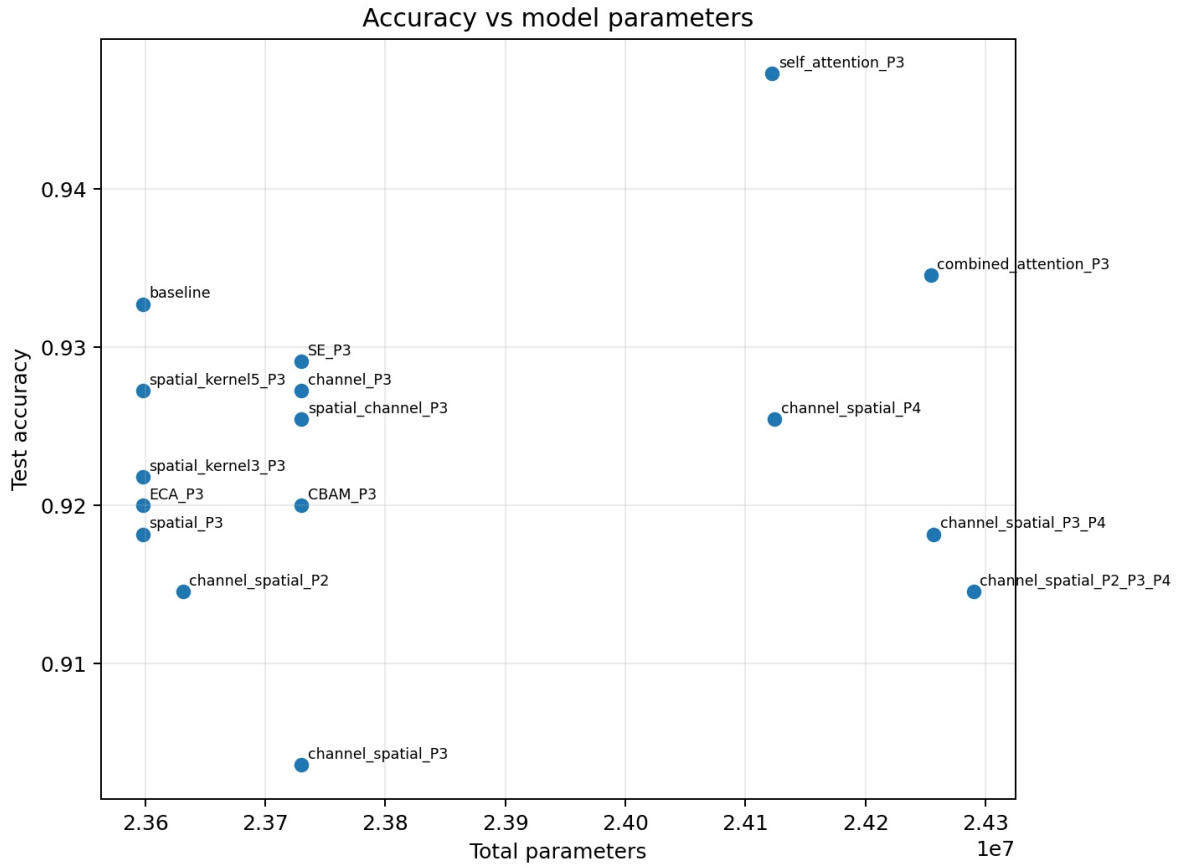
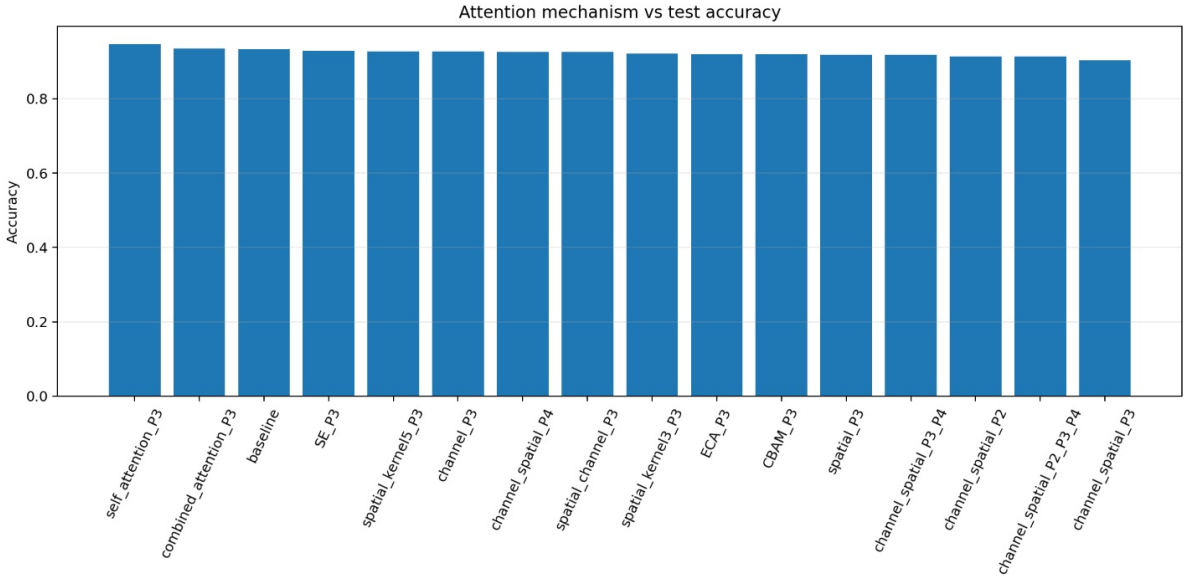
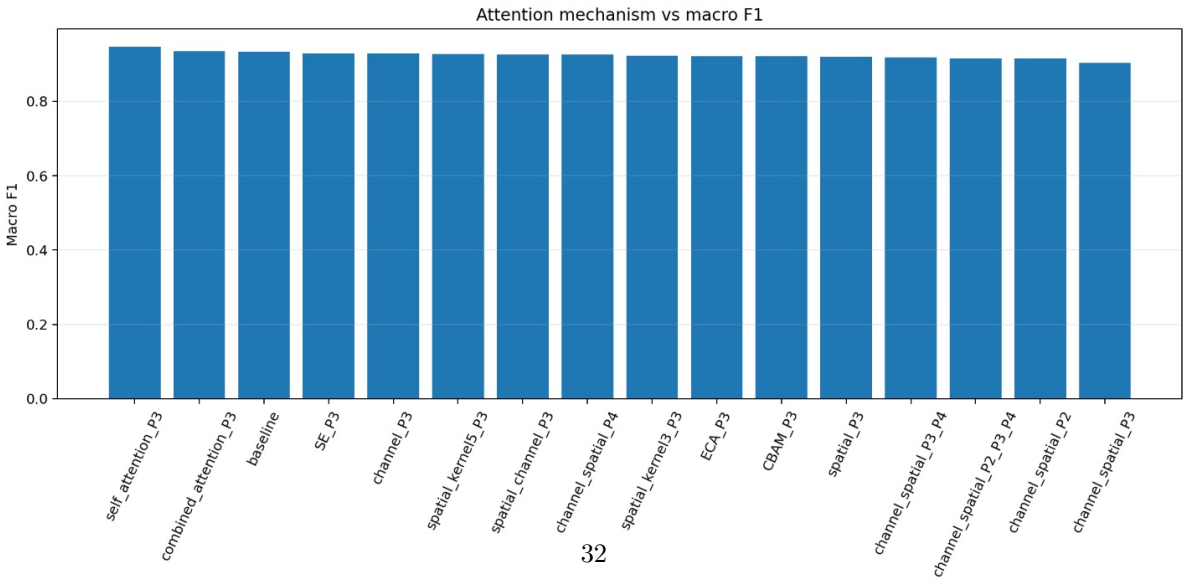


Figure 30: Attention maps at different feature extraction stages: P2, P3 and P4.


Accuracy vs model parameters
Test accuracy
Total parameters
1e7
0.94
0.93
0.92
0.91
2.36
2.37
2.38
2.39
2.40
2.41
2.42
2.43
self_attention_P3
combined_attention_P3
baseline
SE_P3
spatial_kernel5_P3
channel_P3
spatial_channel_P3
channel_spatial_P4
spatial_kernel3_P3
ECA_P3
CBAM_P3
spatial_P3
channel_spatial_P3_P4
channel_spatial_P2
channel_spatial_P2_P3_P4
channel_spatial_P3


Attention mechanism vs test accuracy
Accuracy
0.0
0.2
0.4
0.6
0.8
self_attention_P3
combined_attention_P3
baseline
SE_P3
spatial_kernel5_P3
channel_P3
channel_spatial_P4
spatial_channel_P3
spatial_kernel3_P3
ECA_P3
CBAM_P3
spatial_P3
channel_spatial_P3_P4
channel_spatial_P2
channel_spatial_P2_P3_P4
channel_spatial_P3


Attention mechanism vs macro F1
Macro F1
0.0
0.2
0.4
0.6
0.8
self_attention_P3
combined_attention_P3
baseline
SE_P3
channel_P3
spatial_kernel5_P3
spatial_channel_P3
channel_spatial_P4
spatial_kernel3_P3
ECA_P3
CBAM_P3
spatial_P3
channel_spatial_P3_P4
channel_spatial_P2_P3_P4
channel_spatial_P2
channel_spatial_P3

32. Recommended Visualization Layout

For one representative test image, students are encouraged to generate a figure in the following format:

Input	Baseline Grad-CAM	Channel Grad-CAM	Spatial Attention Map	Channel+Spatial Grad-CAM

For students completing the additional attention comparison, the figure may be extended to:

Input|Baseline|SE|ECA|CBAM|Self-Attention

using appropriate visualization methods.

33. Required Experimental Analysis

The final report should contain three forms of analysis.

33.1 Quantitative Analysis

Report:

Accuracy + Precision + Recall + Macro F1

33.2 Complexity Analysis

Report:

Parameters + Inference Time

33.3 Qualitative Analysis

Report:

Grad-CAM + Attention Maps + Visual Interpretation

Therefore, the overall evaluation should be:

Quantitative + Computational + Qualitative

34. Experimental Protocol

To ensure fair comparison, all ablation experiments must use:

- identical training images;
- identical validation images;
- identical testing images;
- identical random seed where possible;
- identical image preprocessing;
- identical augmentation;
- identical backbone;
- identical classifier;
- identical optimizer;
- identical batch size;
- identical number of epochs;
- identical learning-rate schedule;
- identical fine-tuning procedure; and
- identical evaluation metrics.

Only the experimental factor under investigation should change.
For example, when studying attention position:

Only Attention Position Should Change

and when studying attention type:

Only Attention Type Should Change.

35. Minimum Mandatory Ablation Set

If laboratory time is limited, every student must complete at least the following experiments:

1. Baseline pretrained ResNet50.
2. Channel attention at P_3.
3. Spatial attention at P_3.
4. Channel+Spatial attention at P_3.
5. Channel+Spatial attention at P_2.
6. Channel+Spatial attention at P_4.
7. Self-attention at P_3.
8. Grad-CAM comparison between baseline ResNet50 and one selected attention model.
9. Spatial-attention visualization at P_2, P_3, and P_4.
10. Graphical comparison of the classification performance of the attention models.

This minimum set allows students to study:

Attention Type

and

Attention Position.

36. Additional Student Exploration

Students interested in extended experimentation should select at least two mechanisms from:

SE, ECA, CBAM, Coordinate Attention, Self-Attention

and integrate them with the same ResNet50 backbone.
Students should maintain the same:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;
- epochs;
- fine-tuning strategy; and
- attention position.

The additional investigation should include both:

Quantitative Comparison

and

Qualitative Comparison.

37. Questions Students Must Answer

1. **What is the purpose of residual connections in ResNet50?**
   Residual connections help the network learn deeper features and reduce the vanishing gradient problem.
2. **What does conv2_x represent compared with conv5_x?**
   Conv2_x learns basic features like edges while conv5_x learns complex features like flower shapes.

3. **Why does the spatial resolution decrease with network depth?**
   It decreases to reduce computation and learn more meaningful features.
4. **Why does the number of feature channels increase with network depth?**
   More channels help the network learn different and complex features.
5. **What is meant by low-level, intermediate, and high-level features?**
   low-level features are edges and colors,intermediate features are textures and patterns, and high-level features are object shapes.
6. **What is the purpose of an attention mechanism?**
   attention helps the model focus on important features and ignore less useful information.
7. **What does channel attention attempt to learn?**
   It learns which feature channels are more important for classification.
8. **What does spatial attention attempt to learn?**
   It learns which locations in the feature map are more important.
9. **How is channel attention different from spatial attention?**
   Channel attention focuses on feature channels while spatial attention focuses on image locations.
10. **Why can channel and spatial attention be used sequentially?**
    Channel attention selects useful features first and spatial attention then focuses on important locations.
11. **Why might channel attention behave differently at conv3_x and conv5_x?**
    conv3_x contains intermediate features while conv5_x contains more complex features so the important channels may differ.
12. **Why can spatial attention be useful at an intermediate ResNet stage?**
    It can focus on important regions while the feature map still contains good spatial details.
13. **Why is self-attention computationally expensive when the feature map has high spatial resolution?**
    It compares many spatial positions with each other which needs more computation.
14. **What happens to the size of the self-attention matrix when the spatial resolution increases?**
    The matrix becomes much larger because it grows with the square of the number of spatial positions.
15. **How does attention performance vary when the insertion position is changed?**
    Performance can change because different stages learn different types of features.
16. **How does attention at P_2 differ visually from attention at P_4?**
    P_2 can show finer details, while P_4 focuses more on larger and complex patterns.
17. **Does applying attention at multiple stages always improve classification performance?**
    no,sometimes it can increase computation or reduce accuracy.
18. **Does Channel→Spatial attention behave differently from Spatial→Channel attention?**
    yes,the order can affect the features selected by the model and its final performance.
19. **How does the spatial-attention kernel size influence the attention map?**
    A small kernel focuses on local details while a larger kernel captures wider spatial information.
20. **How does SE attention differ from ECA attention?**
    SE uses fully connected layers to learn channel weights while ECA uses a small one-dimensional convolution.
21. **How does CBAM combine channel and spatial attention?**
    CBAM applies channel attention followed by spatial attention.
22. **How does self-attention differ from channel attention?**
    self attention relates different spatial positions while channel attention gives importance to feature channels.
23. **Does the attention mechanism with the largest number of parameters necessarily provide the highest accuracy?**
    no,more parameters do not always mean better accuracy.
24. **What is the difference between an internal attention map and Grad-CAM?**
    An attention map shows learned attention weights while Grad-CAM uses gradients to highlight regions important for a prediction.
25. **Why is Grad-CAM called a post-hoc explanation technique?**
    Because it explains the model's prediction after the model has already made it.
26. **Does the attention model suppress irrelevant background regions?**
    It can reduce the importance of irrelevant background regions,but it may not remove them completely.
27. **Are the attention maps of correctly classified and incorrectly classified images different?**
    yes,they can differ because the model may focus on different regions or misleading background features.
28. **Which flower classes benefit most from attention?**
    This depends on the experimental results.The classes with improved class wise performance benefit more.
29. **What information can be obtained from a confusion matrix that cannot be obtained from overall accuracy alone?**
    It shows which classes are confused with each other and the number of correct and incorrect predictions.
30. **Why is macro F1 useful in addition to accuracy?**
    macro F1 calculates the average F1 score across classes so we can understand performance on each class better.
31. **What is meant by the accuracy–complexity trade-off?**

It means balancing better accuracy with the number of parameters and computation required.

32. **What is meant by the F1–inference-time trade-off?**
    It means balancing good classification performance with how quickly the model makes predictions.
33. **Why must every ablation study use the same dataset split?**
    The same split makes the comparison fair because every model is tested on the same data.
34. **Why must the same training protocol be maintained when comparing different attention mechanisms?**
    It ensures that the performance difference is mainly due to the attention mechanism and not different training settings.
35. **What conclusions can be drawn from the Grad-CAM comparison?**
    It shows whether the models focus on the flower regions or on irrelevant background areas when making predictions.

38. Observations to be Recorded

Students should record the following during experimentation:

1. Input and output feature dimensions for each attention module.
2. Training accuracy.
3. Validation accuracy.
4. Testing accuracy.
5. Training loss.
6. Validation loss.
7. Macro precision.
8. Macro recall.
9. Macro F1-score.
10. Number of trainable parameters.
11. Total number of model parameters.
12. Average inference time.
13. Confusion matrix.
14. Channel-attention weights.
15. Spatial-attention maps.
16. Self-attention responses.
17. Grad-CAM maps.
18. Attention maps for different insertion positions.

39. Suggested Laboratory Report Structure

The laboratory report may be organized as follows:

1. **Aim**
2. **Dataset Description**
3. **ResNet50 Architecture**
4. **Attention Mechanisms**
5. **Proposed Experimental Architecture**
6. **Experimental Setup**
7. **Attention-Type Ablation**
8. **Attention-Position Ablation**
9. **Performance Results**
10. **Complexity Analysis**
11. **Training and Validation Graphs**
12. **Confusion Matrix Analysis**
13. **Grad-CAM Analysis**
14. **Internal Attention-Map Analysis**
15. **Qualitative Comparison**
16. **Additional Attention-Mechanism Comparison**
17. **Discussion**
18. **Conclusion**

40. Discussion Guidelines

Students should not merely reproduce numerical results.

The discussion should explain:

- why a particular attention mechanism produced a change in performance;
- how attention location influences feature refinement;
- how feature resolution changes attention behavior;
- how model complexity changes with different mechanisms;
- whether performance improvements justify additional computational cost;
- whether attention maps correspond to meaningful object regions;
- whether attention helps suppress the background;
- whether attention behaves differently for incorrect predictions; and
- whether quantitative and qualitative observations support each other.

Students should avoid making conclusions based on a single image.

Qualitative observations should preferably be based on several representative test images.

Inference from the Experiment

self attention at P3 achieved the highest accuracy of 94.73% and macro F1 score of 94.74% which is better than the baseline model.This may be because self attention helps the model to learn relationships between different image regions but all attention mechanisms did not improve the accuracyas channel spatial attention at P3 achieved only 90.36%.

The position of attention also affects the performance P2 focuses more on small details while P3 and P4 learn deeper features.From the results we can see that applying attention at different positions gives different accuracy.Also applying attention at multiple stages did not always improve the performance.

The attention mechanisms increase the number of parameters by different amounts.The baseline model has 23.59 million parameters while the self-attention model has 24.12 million parameters but self attention achieved better accuracy with less measured inference time than the baseline so it gives a good trade off in this experiment.

From the attention maps and GradCAM images we can observe which regions are getting more importance during prediction.These visualizations help to check whether the model focuses on the flower or the background.incorrect predictions may happen when the model focuses on less useful regions or similar features between flower classes.

the numerical results and visualizations help us to understand the model performance better.self attention at P3 performed best among the tested models but the attention maps should also be checked on multiple images before making a final conclusion.

41. Expected Outcome

At the end of the experiment, students should understand that an attention mechanism is not an isolated module whose behavior is independent of the backbone.

Its effectiveness depends on:

$$\begin{aligned}\text{Attention Effectiveness} = {} & \text{Attention Type} \\ & + \text{Attention Position} \\ & + \text{Feature Resolution} \\ & + \text{Feature Semantics} \\ & + \text{Model Complexity.}\end{aligned}$$

Students should also understand that classification performance alone does not provide a complete explanation of an attention mechanism.

A comprehensive analysis should therefore combine:

$$\boxed{\text{Classification Performance}}$$

with

$$\boxed{\text{Computational Complexity}}$$

and

$$\boxed{\text{Qualitative Visualization.}}$$

Therefore:

$$\boxed{\text{Complete Attention Analysis} = \text{Quantitative} + \text{Computational} + \text{Qualitative}}$$

The final conclusions must be based on the student's experimental results rather than on assumptions about which attention mechanism should perform best.

42. References

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0.1 GitHub Repository

You can access the source code through this link. **GitHub Link:** https://github.com/Nitheesha-1025/Deep-Learning/tree/Effect-of-Attention-Type-and-Attention-Position